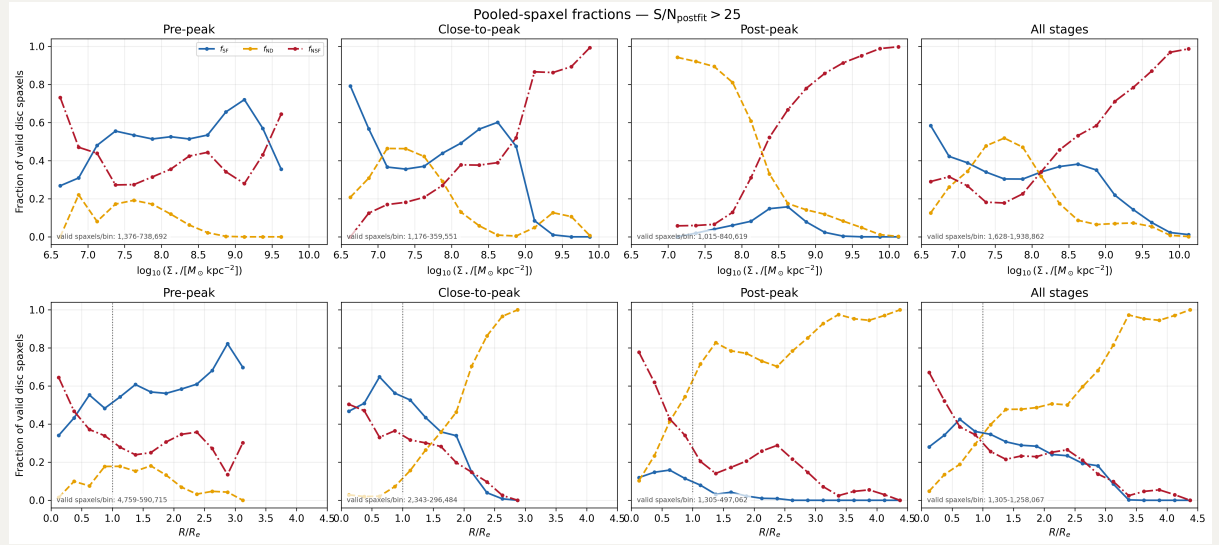


20260906 Where Star Formation Remains along Infall Stages

First of all, instead of a cut at stellar mass and radius, we perform a cut at `POSTFIT_SNR>25` to remove those faint, unreliable spaxels. We also perform a equal-galaxy plots, and the trends are similar.



In general, outer low- $\Sigma_\star$ regions mainly become ND, while central high- $\Sigma_\star$ regions remain emission-line detected but are increasingly NSF:

1. Pre-peak to close-to-peak: outside-in erosion. The decline in f_{SF} is larger at $1 - 2R_e$ than inside R_e , and this outer loss comes with increasing f_{ND} .
2. Close-to-peak to post-peak: a large loss of SF regions across the observed disc. NSF becomes dominant within $0.5R_e$ and ND dominates outside.
3. Therefore, the full pre-to-post difference is stronger quenching outside but then substantial everywhere. Outside-in quenching plus disc-wide reduction is more accurate than either pure outside-in or uniform quenching.

If we look at $f_{\text{SF}}(R/R_e)$, we will find in pre-peak sample, it is $0.4 \sim 0.8$ with slightly positive slope across the disc up to $3R_e$. And in close-to-peak, its central value remains 0.5 and gradually declines and reaches 0 at $3R_e$. Then in post-peak, it is only 0.1 to 0.2 within R_e and almost 0 SF outside $1.5R_e$. For $f_{\text{NSF}}(R/R_e)$, it shows steeper decrease along infall stages and dominates within $0.5R_e$ in post-peak. Conversely, $f_{\text{ND}}(R/R_e)$ becomes steeper increase in close-to-peak and post-peak, while only $\lesssim 0.2$ in all radius in pre-peak.

Σ_* tends also give a slightly different prospective. In pre-peak, ND is $\lesssim 0.2$ and SF dominantes from 7.0 to 9.5. In close-to-peak, f_{ND} rises especially in low-mass end, while the sharp SF drop in $\log(\Sigma_*) > 9.0$ goes to NSF. Then in post-peak, only $\lesssim 20\%$ regions still SF, and concentrate between 7.5-9.0. ND dominates <8.25 and NSF dominates >8.25 . And this 8.25 in stellar mass transition seems to correlate with trainsition at $0.6R_e$.

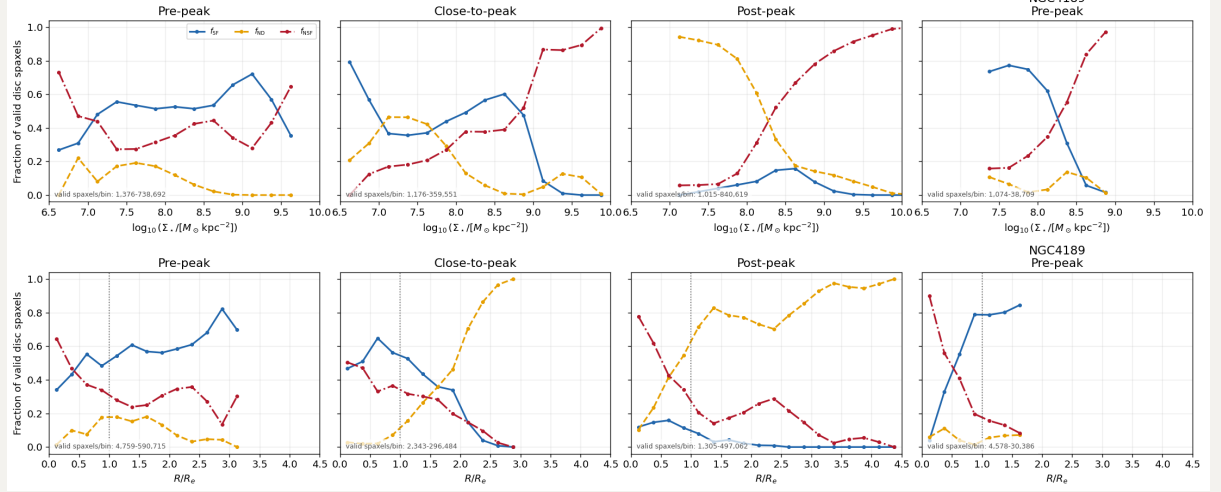
SF becoming ND in outer low- Σ_* has a natural RPS interpretation. Since those regions have lower Σ_* and Σ_{gas} , and so a shallower local gravitational potential/restoring force, then the outside-in gas removal removes the conditions required to maintain H II regions. Therefore, rather than changing the excitation mechanism of the outer gas, those outer low- Σ_* regions lose sufficiently strong nebular emission. The pre-peak to close-to-peak transition further support this explanation. The close-to-peak samples still maintain $f_{\text{SF}} \gtrsim 0.4$ up to $2R_e$, but quickly drop to near 0 at $2.3R_e$.

SF becoming NDF in inner high- Σ_* means the presence of ionized gas but the decrease of O/B star dominance. This can be ionization due to shocks/outflows, AGN, DIG, older stellar population, or bulge/nuclear radiation. And we can do a simple classification based on their morphology. For those extraplanar NSF regions (like NGC4064), we know those are starburst outflows; for those NSF regions surround or bridge SF regions (like IC3392), they should be DIG or leaked OB photons; and for those central NSF regions (like NGC4698), they are likely old-star LINER and/or AGN; also there are maybe mixed situations (like NGC4580).

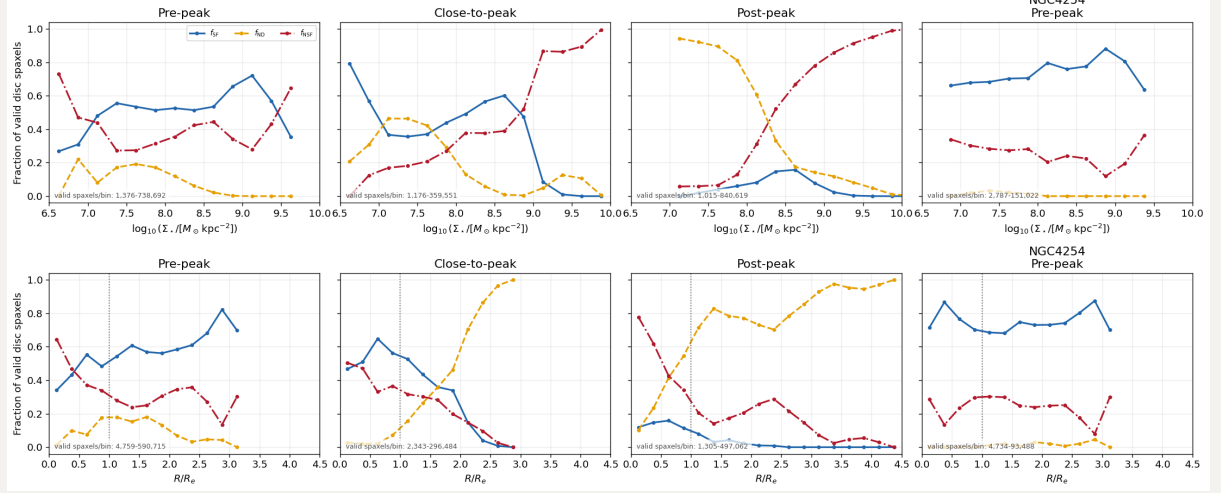
One interesting thing is that in pre-peak and post-peak, both stellar and radial trends show consistent information, which follows the radial profile of stellar mass, but the close-to-peak samples do not act as a clear transition and behave differently in two trends. It only shows the absolute dominance of ND in outer radius, but not necessarily in low mass end; while it only shows the absolute dominance of NSF in high mass end but still relatively higher fraction of SF in inner disc. I think this may due to the fact that our close-to-peak samples are not representative enough (only 5 galaxies here after inclination selection). But i think this may also suggest that ND appears more directly linked to stripping geometry, whereas NSF appears more linked to the local stellar/ionization population. This may also confirm that there is no such a clear intermediate transition phase from pre-peak to post-peak. Iin fact, the spatial maps of close-to-peak samples also support this: some close-to-peak galaxies remain blue-rich with orange ND stay mainly at their edges (NGC4298, NGC4501 and NGC4654), while others have compact central SF embedded in NSF and surrounded by ND (NGC4424 and NGC4694).

Individual Galaxy Trends

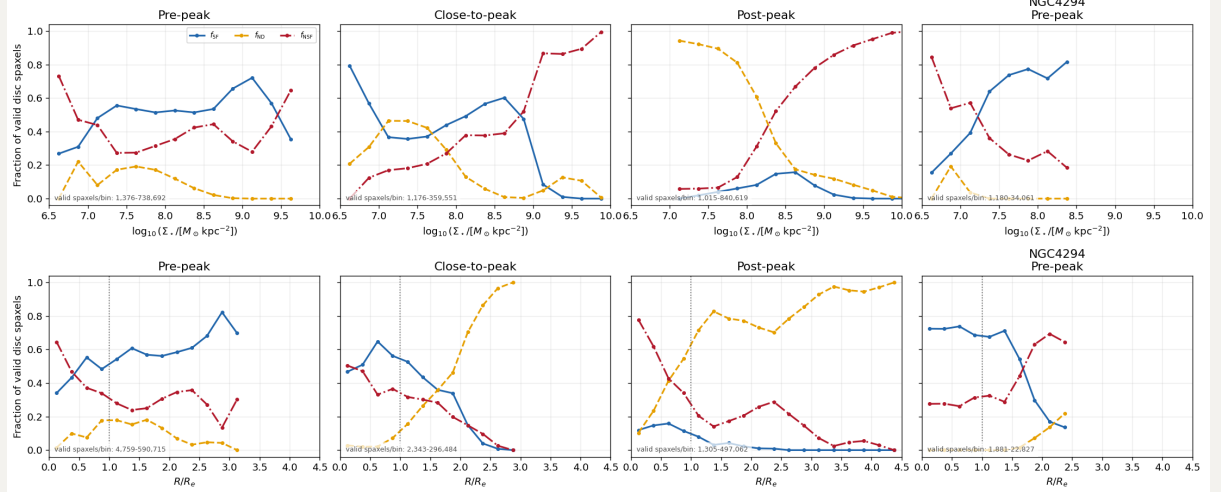
Pooled-spaxel stage references + NGC4189 — $S/N_{\text{postfit}} \geq 25$



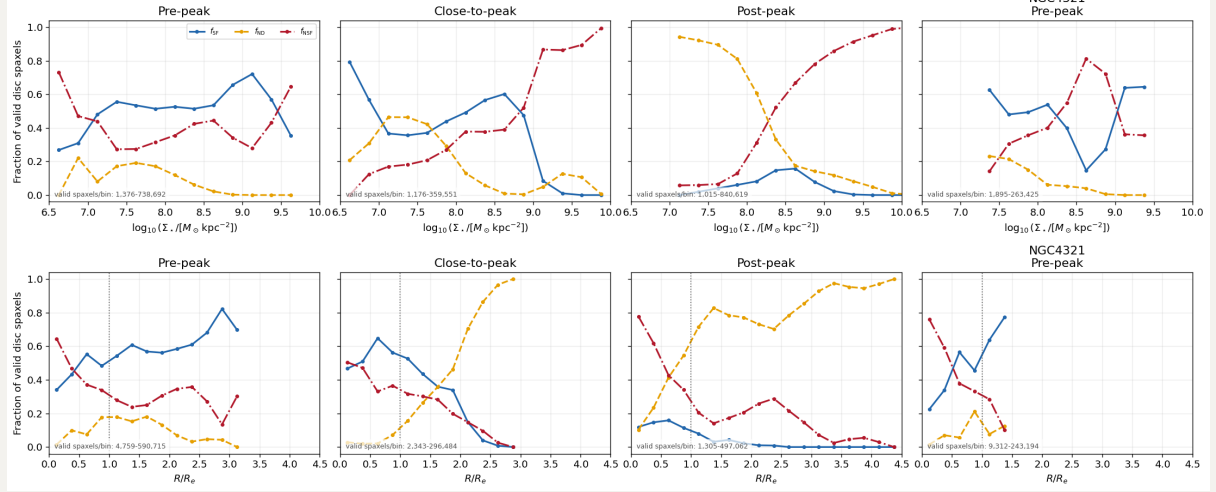
Pooled-spaxel stage references + NGC4254 — $S/N_{\text{postfit}} \geq 25$



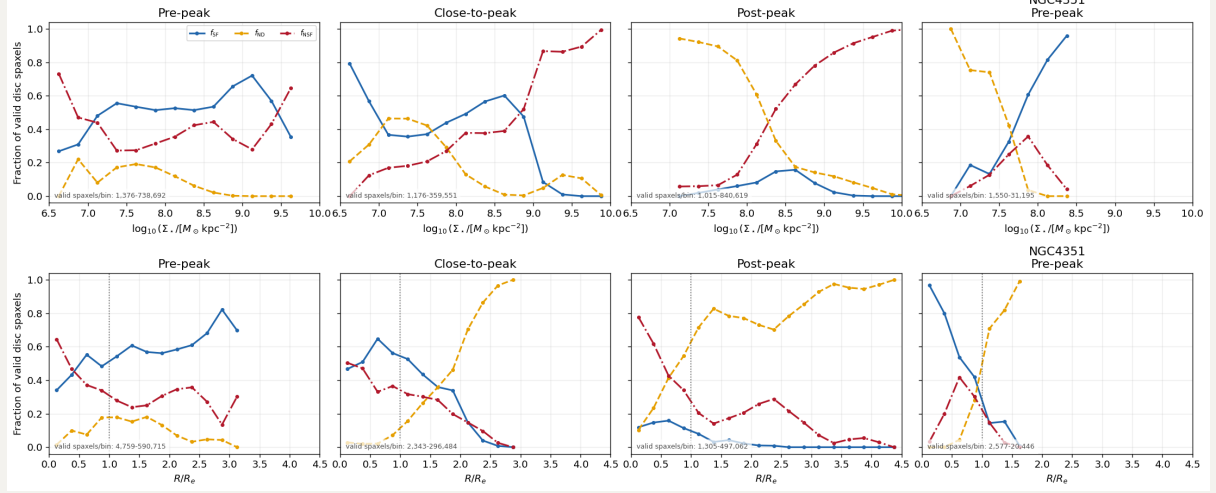
Pooled-spaxel stage references + NGC4294 — $S/N_{\text{postfit}} \geq 25$



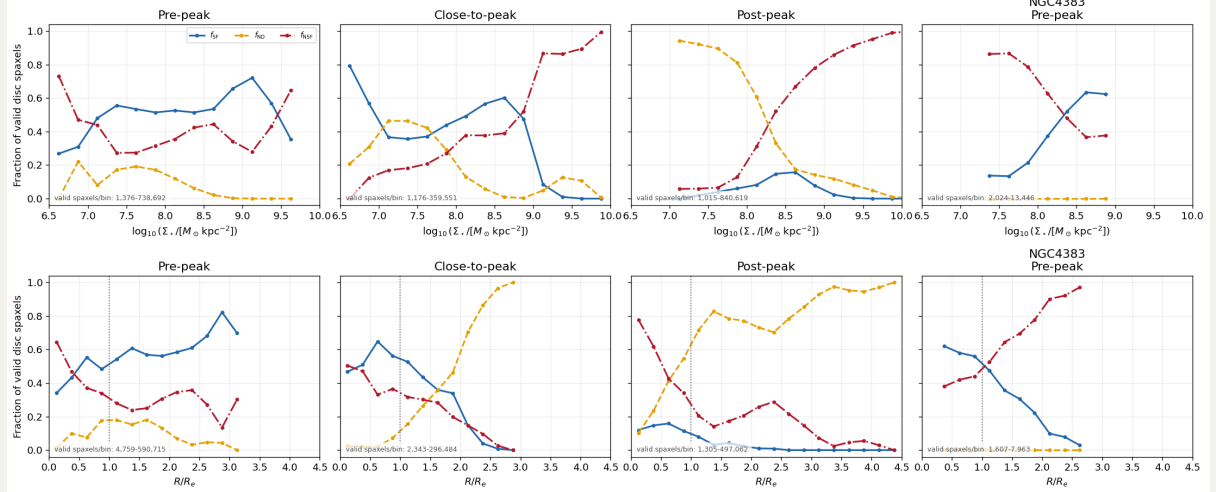
Pooled-spaxel stage references + NGC4321 — $S/N_{\text{postfit}} \geq 25$



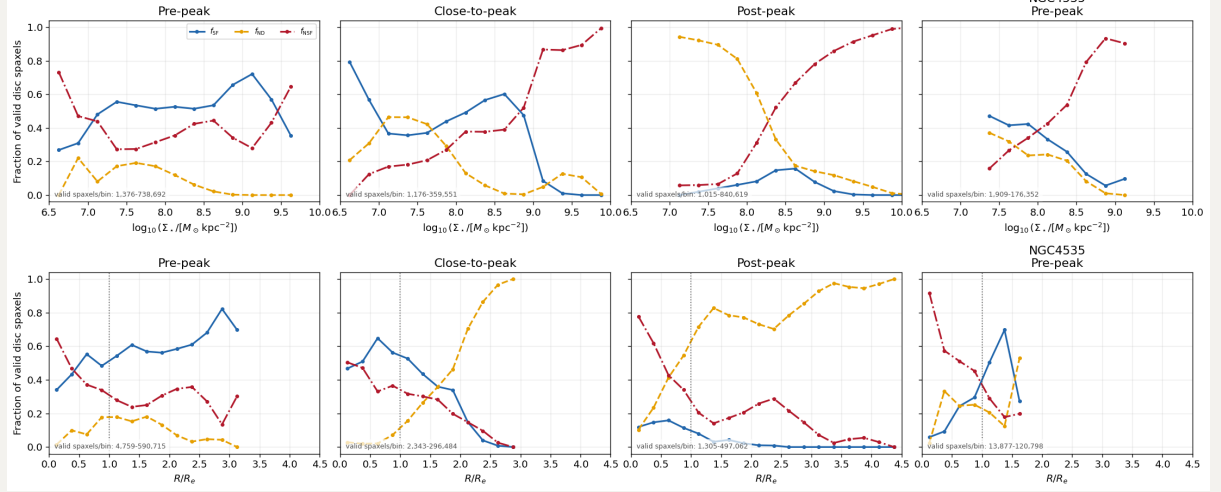
Pooled-spaxel stage references + NGC4351 — $S/N_{\text{postfit}} \geq 25$



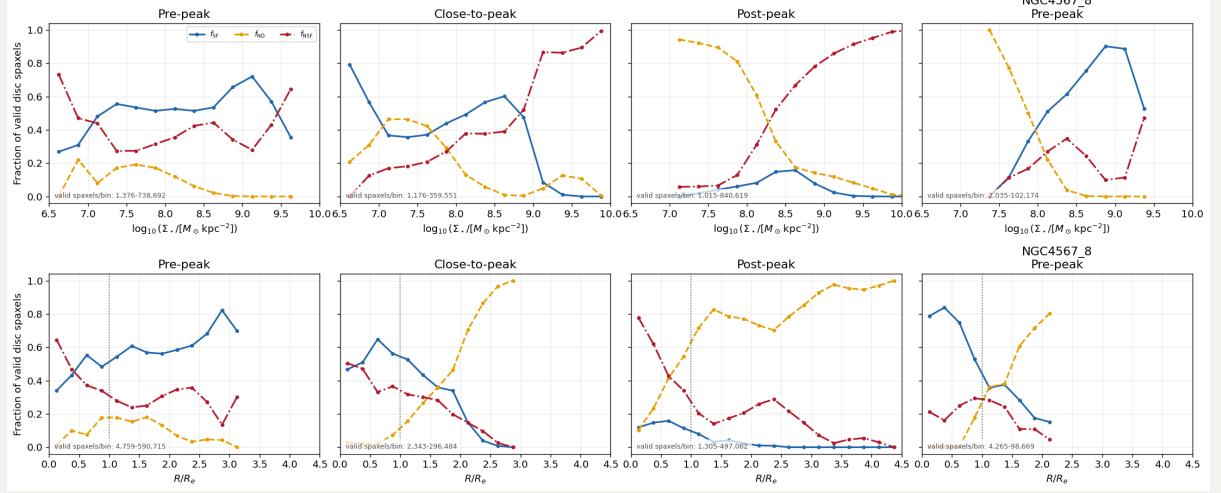
Pooled-spaxel stage references + NGC4383 — $S/N_{\text{postfit}} \geq 25$



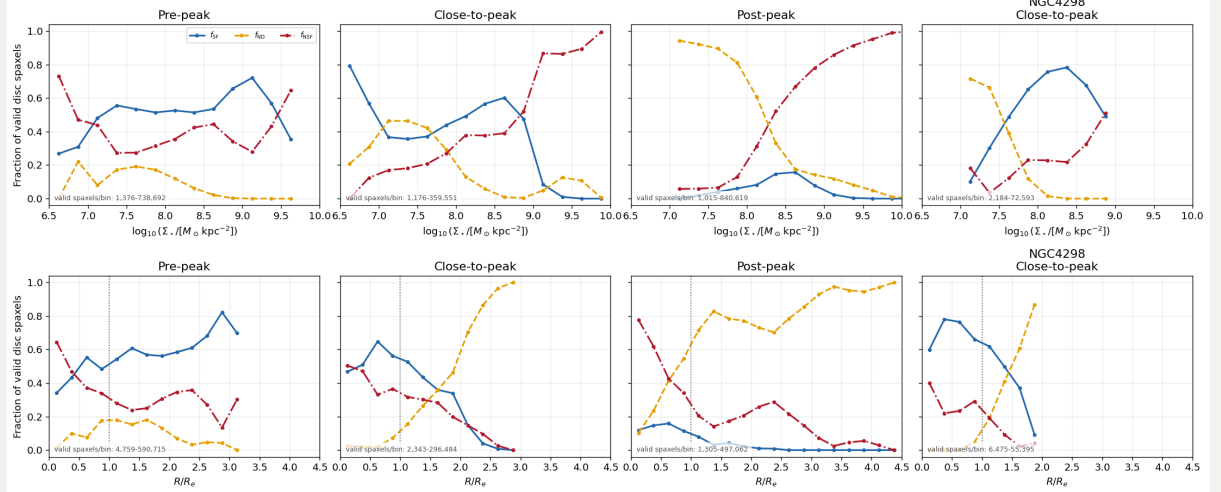
Pooled-spaxel stage references + NGC4535 — $S/N_{\text{postfit}} \geq 25$



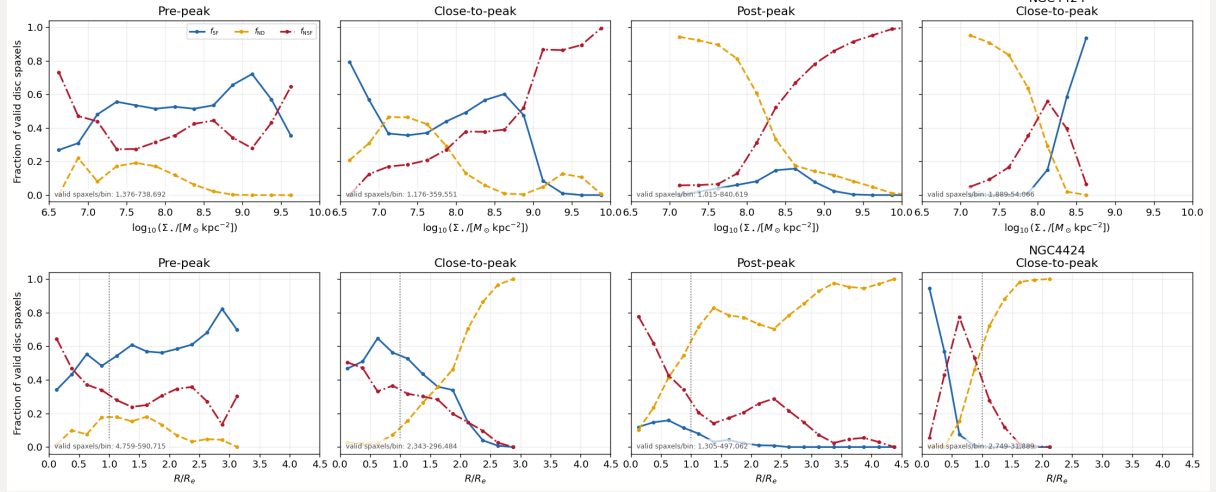
Pooled-spaxel stage references + NGC4567_8 — $S/N_{\text{postfit}} \geq 25$



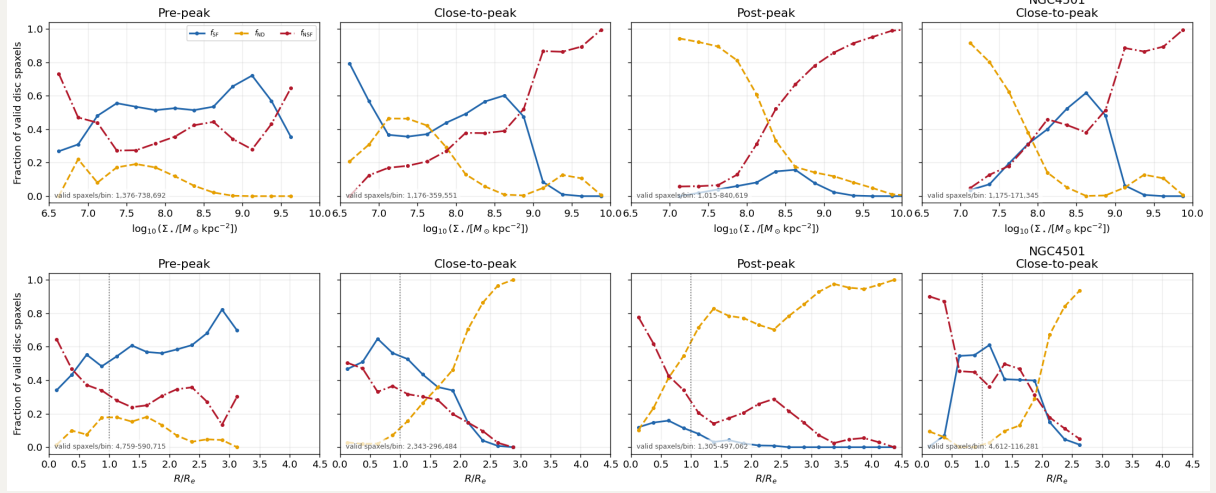
Pooled-spaxel stage references + NGC4298 — $S/N_{\text{postfit}} \geq 25$



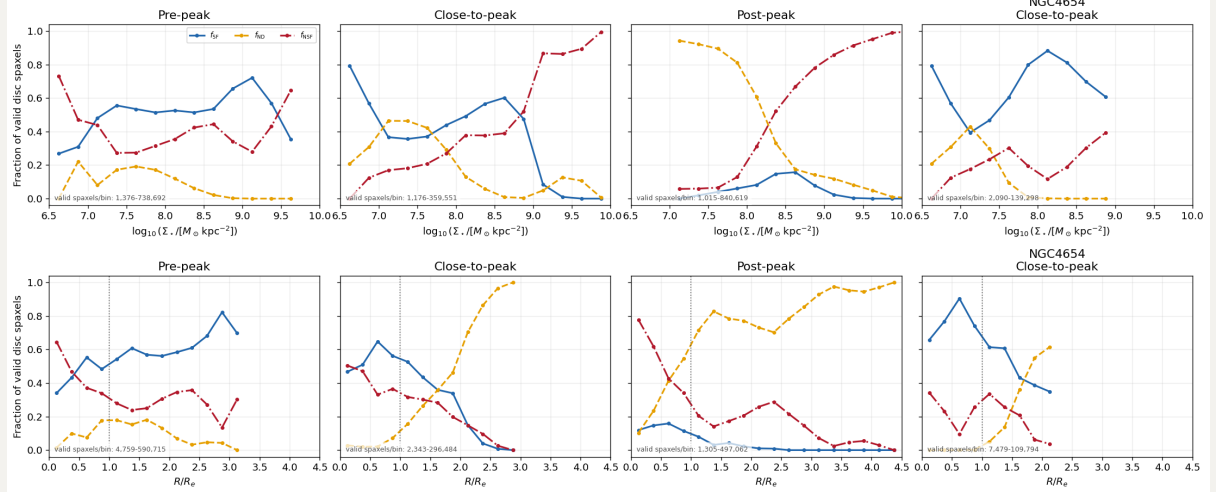
Pooled-spaxel stage references + NGC4424 — $S/N_{\text{postfit}} \geq 25$



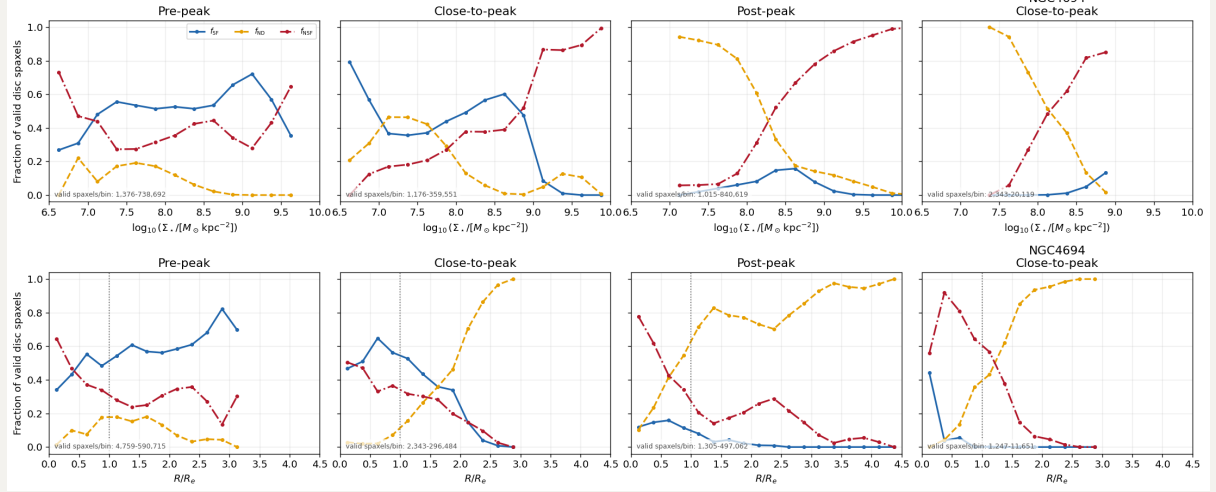
Pooled-spaxel stage references + NGC4501 — $S/N_{\text{postfit}} \geq 25$



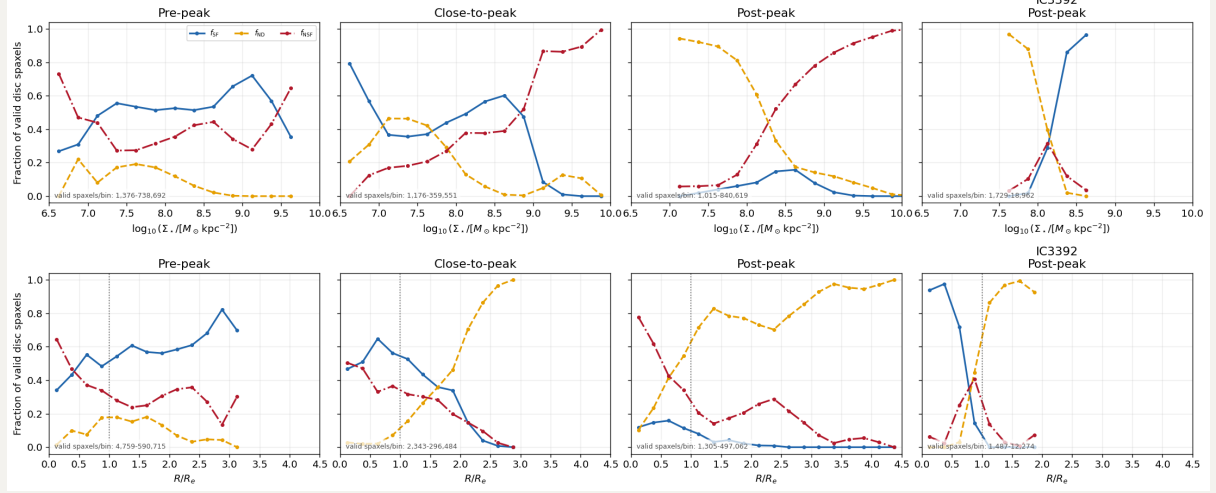
Pooled-spaxel stage references + NGC4654 — $S/N_{\text{postfit}} \geq 25$



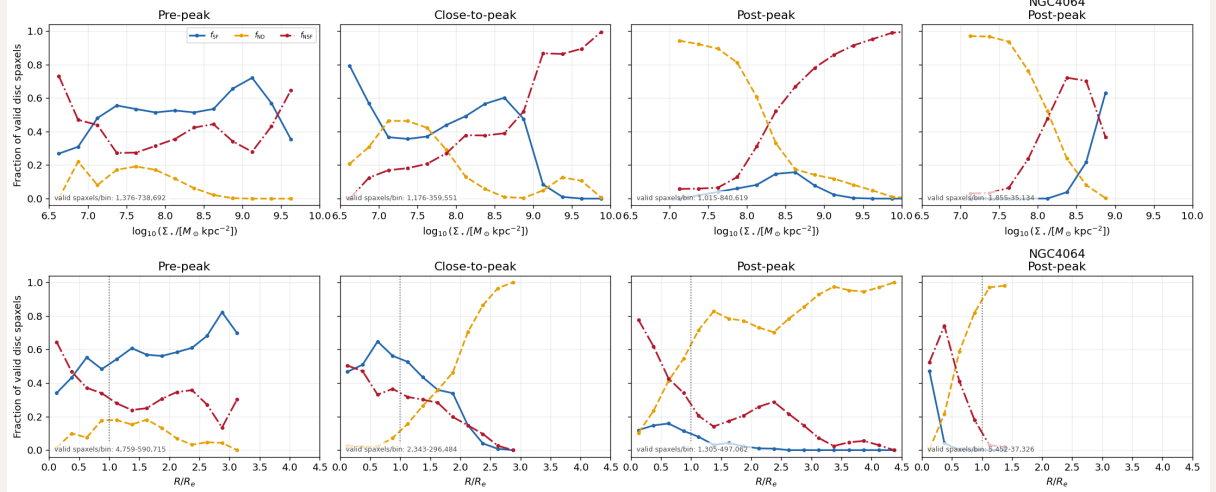
Pooled-spaxel stage references + NGC4694 — $S/N_{\text{postfit}} \geq 25$



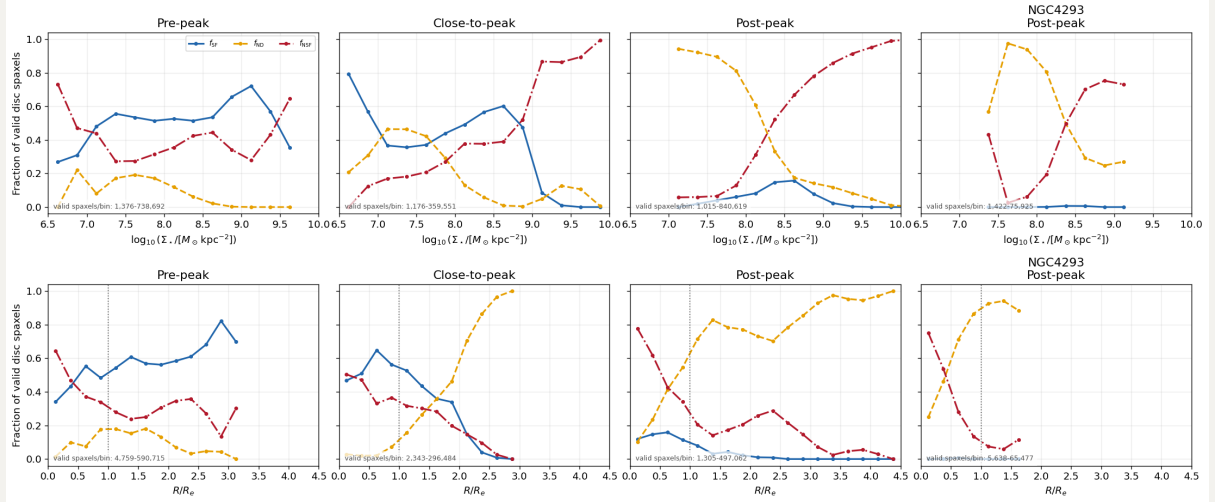
Pooled-spaxel stage references + IC3392 — $S/N_{\text{postfit}} \geq 25$



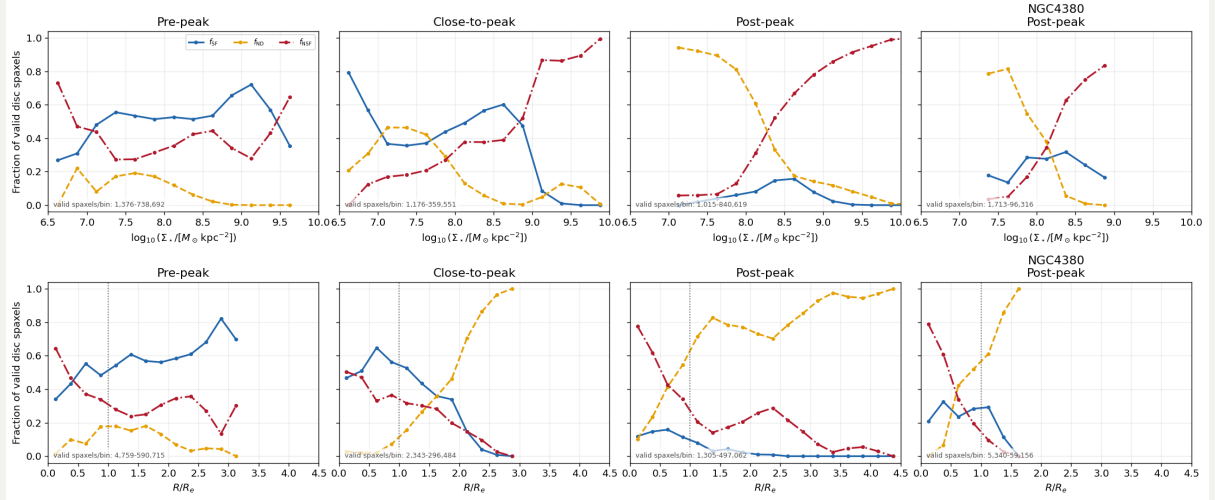
Pooled-spaxel stage references + NGC4064 — $S/N_{\text{postfit}} \geq 25$



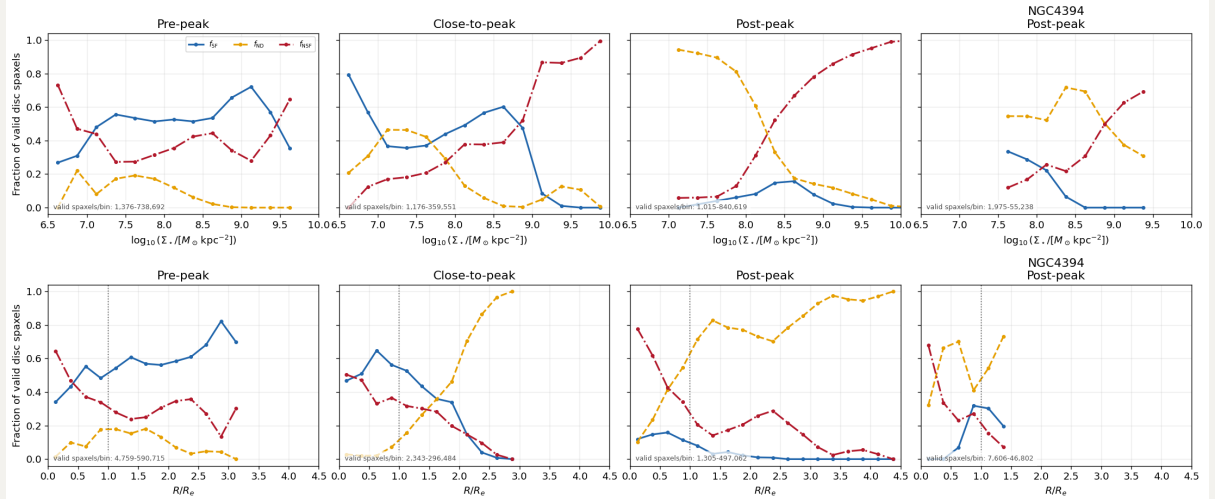
Pooled-spaxel stage references + NGC4293 — $S/N_{\text{postfit}} \geq 25$



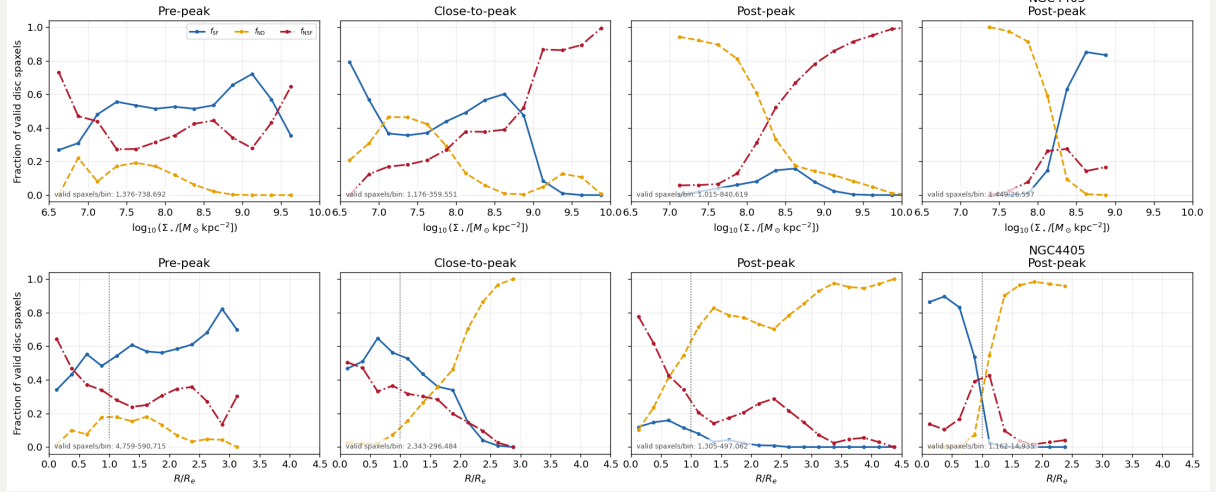
Pooled-spaxel stage references + NGC4380 — $S/N_{\text{postfit}} \geq 25$



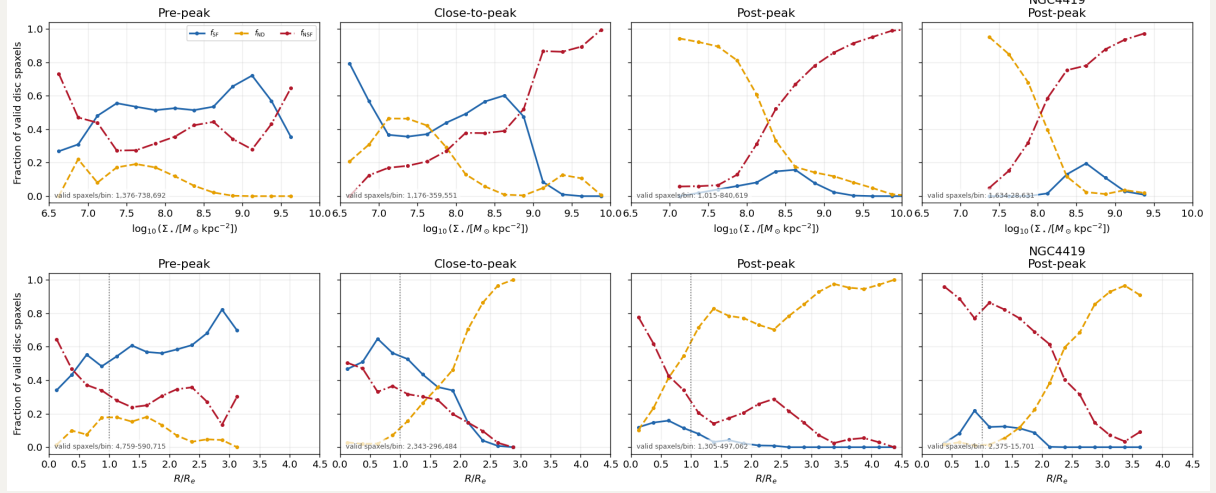
Pooled-spaxel stage references + NGC4394 — $S/N_{\text{postfit}} \geq 25$



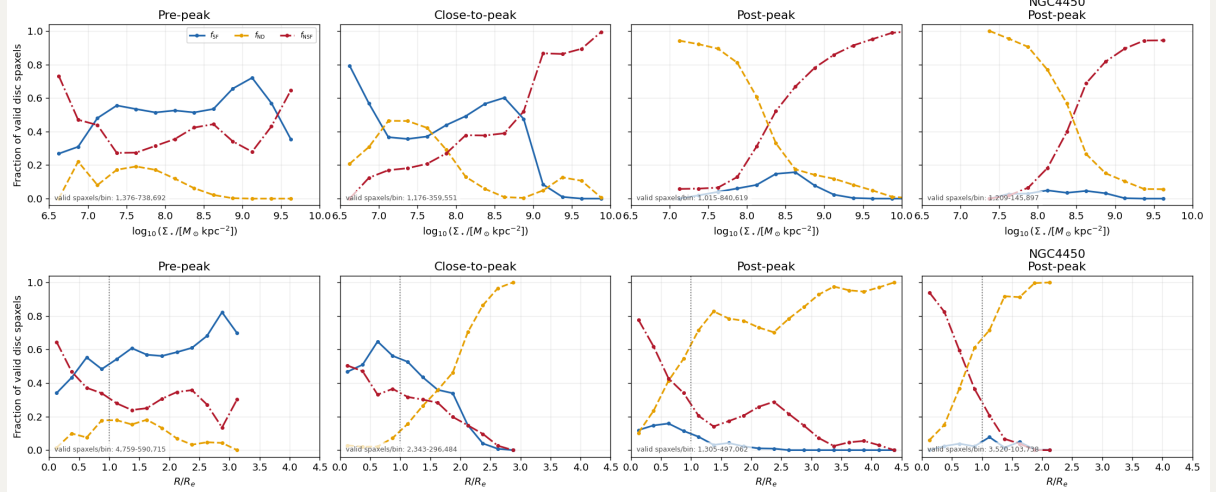
Pooled-spaxel stage references + NGC4405 — $S/N_{\text{postfit}} \geq 25$



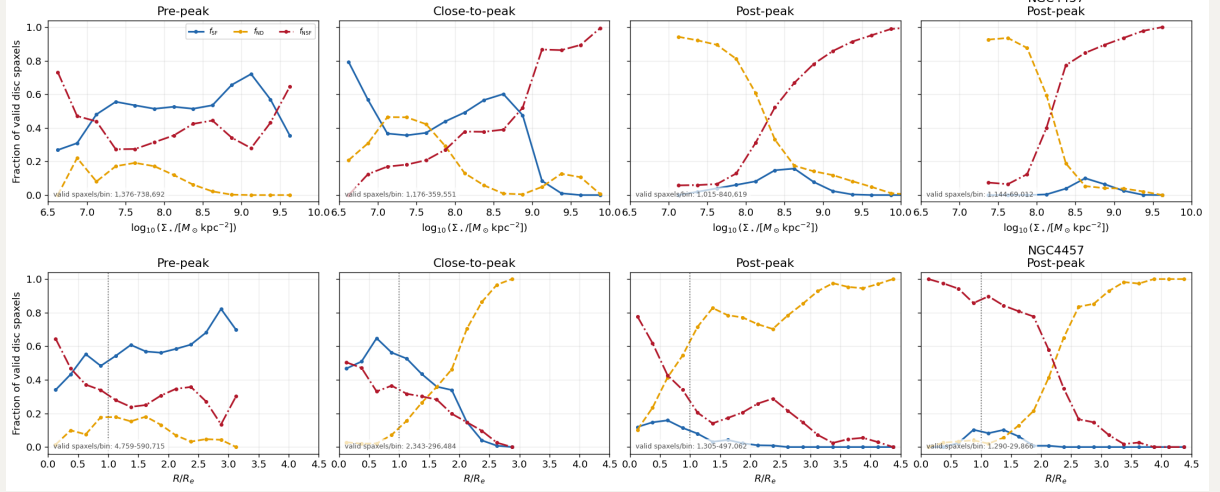
Pooled-spaxel stage references + NGC4419 — $S/N_{\text{postfit}} \geq 25$



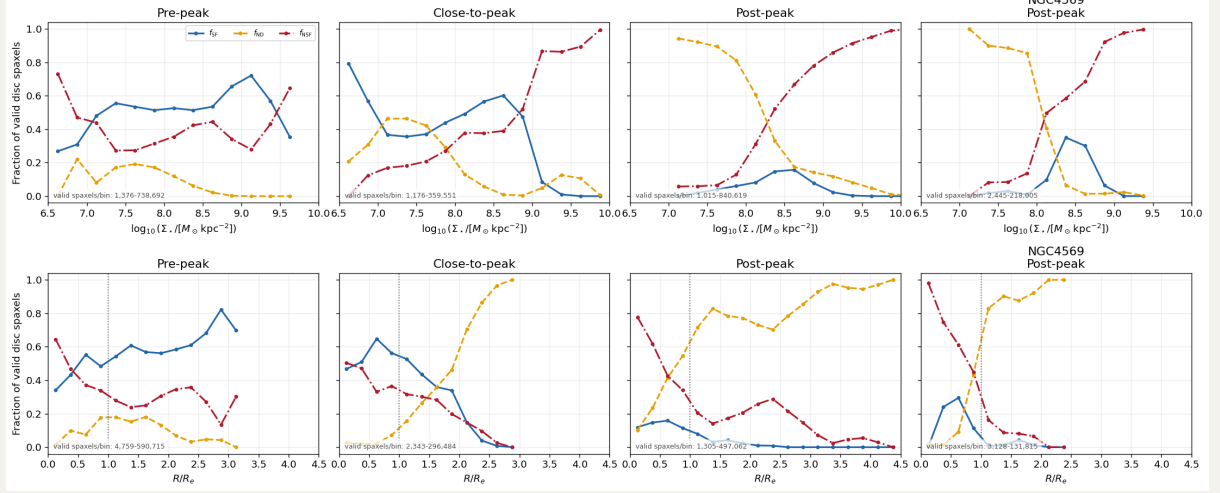
Pooled-spaxel stage references + NGC4450 — $S/N_{\text{postfit}} \geq 25$



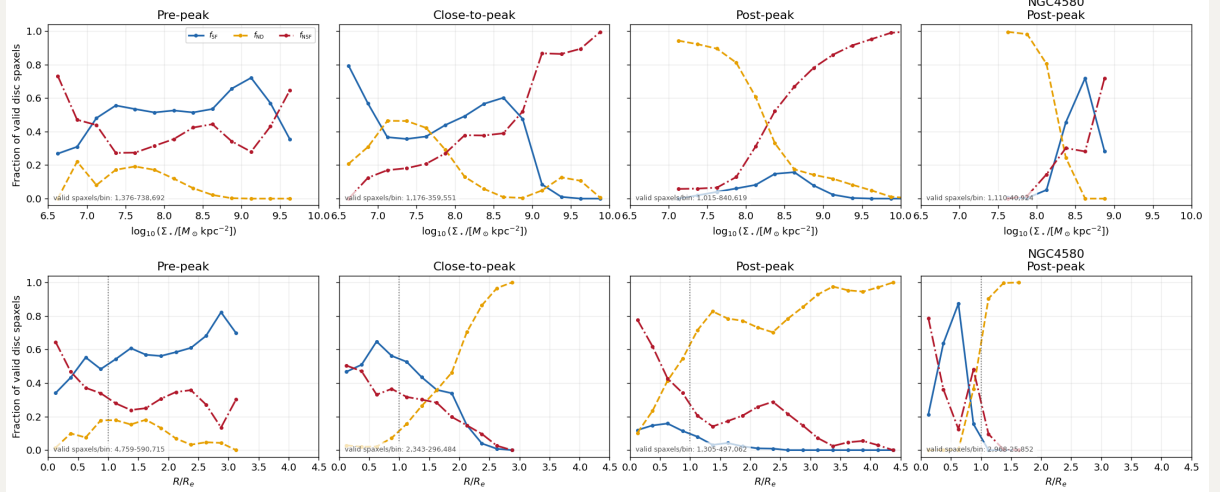
Pooled-spaxel stage references + NGC4457 — $S/N_{\text{postfit}} \geq 25$

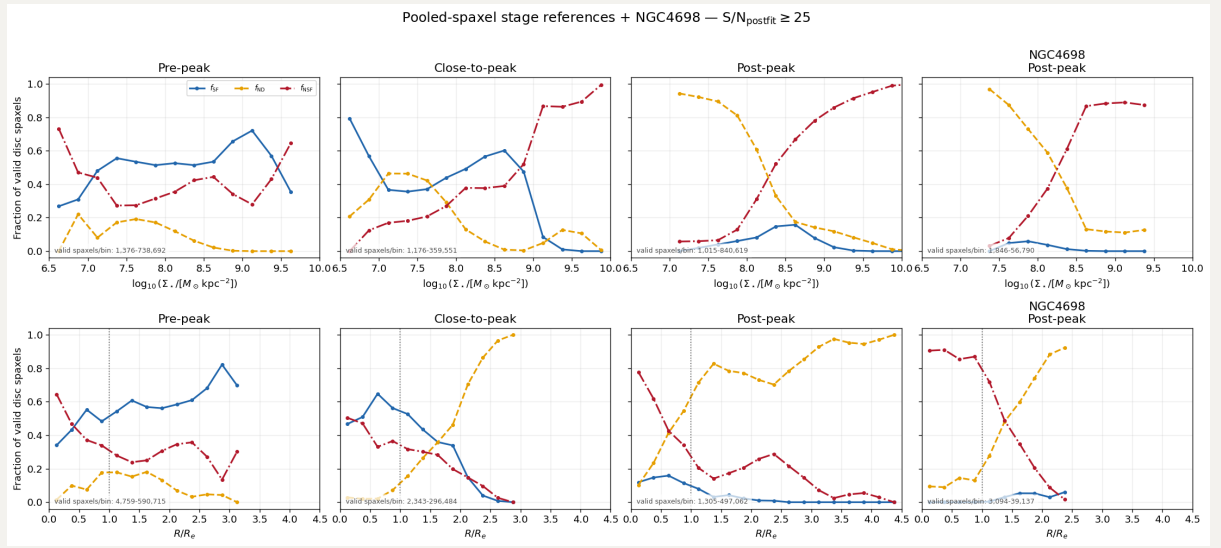
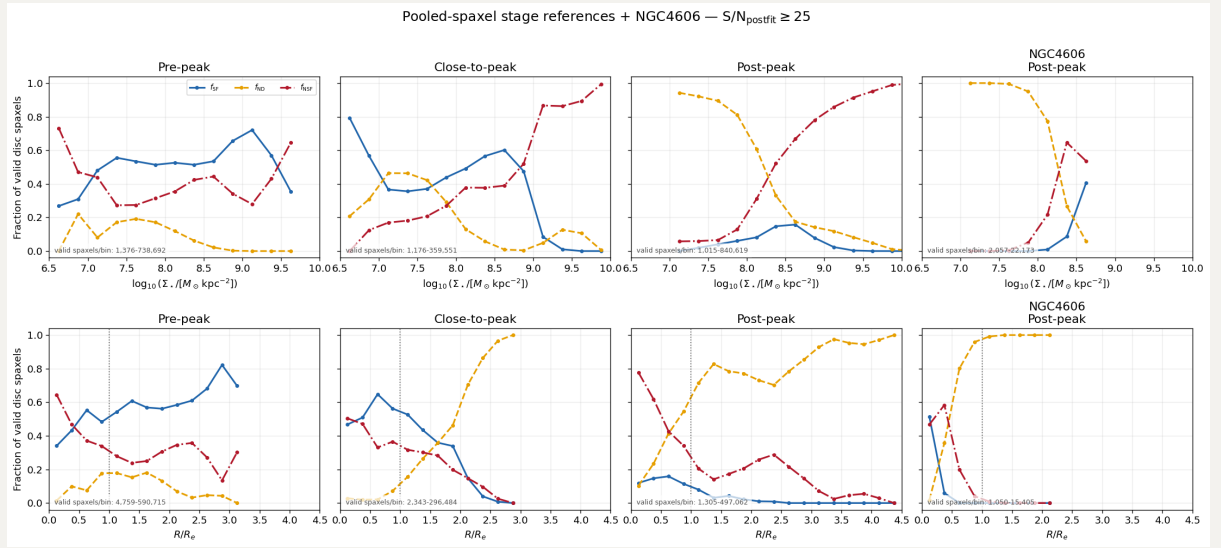


Pooled-spaxel stage references + NGC4569 — $S/N_{\text{postfit}} \geq 25$

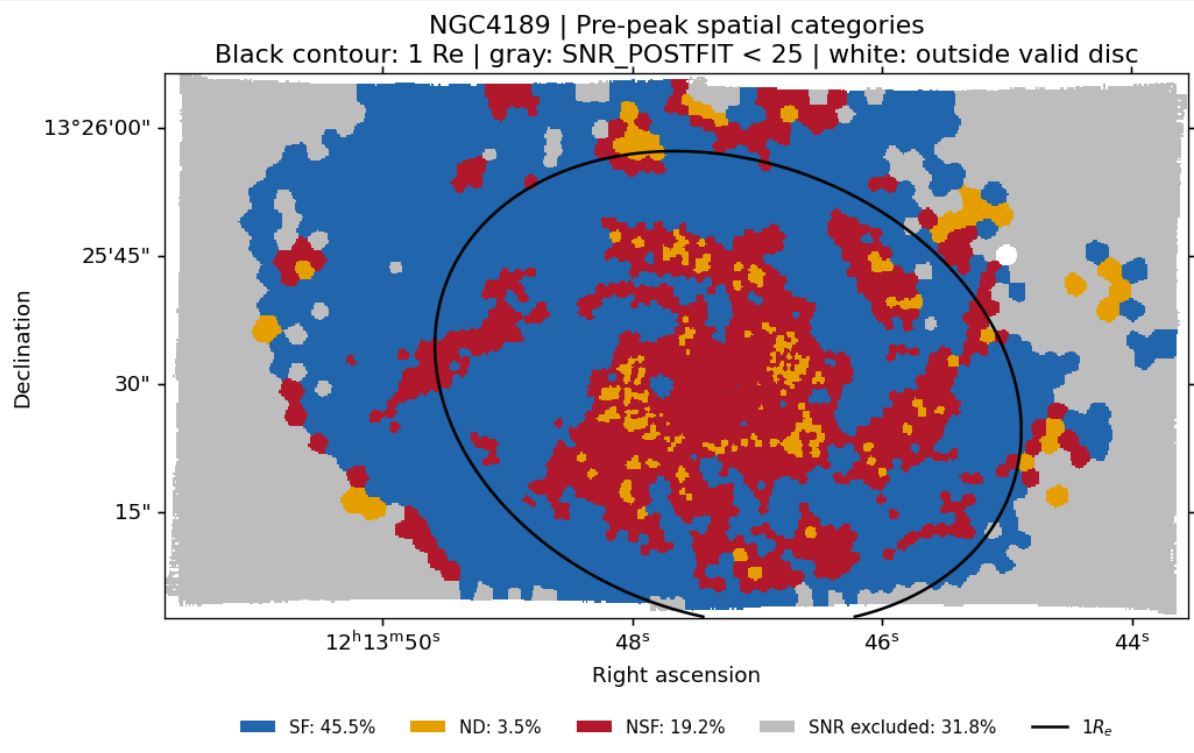


Pooled-spaxel stage references + NGC4580 — $S/N_{\text{postfit}} \geq 25$

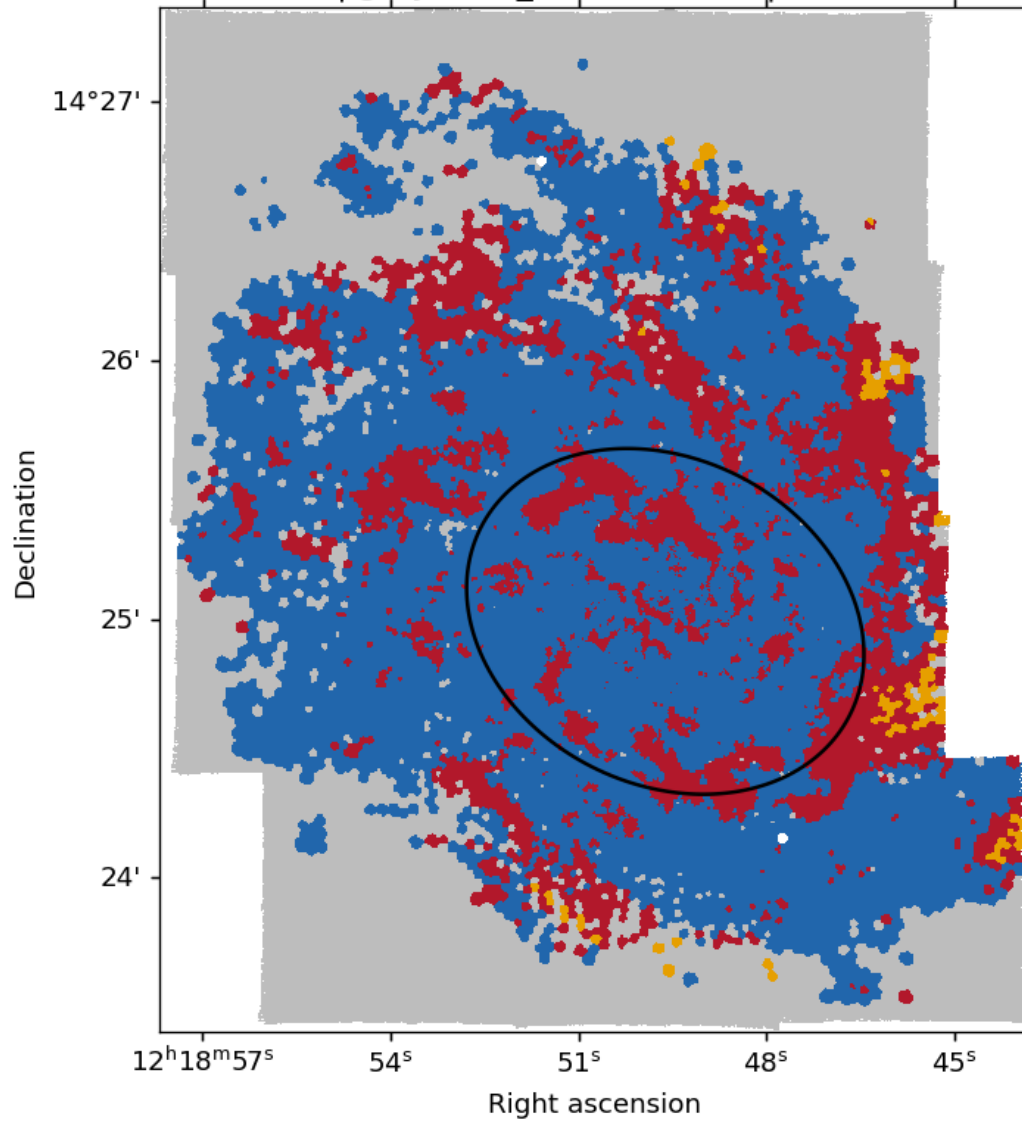




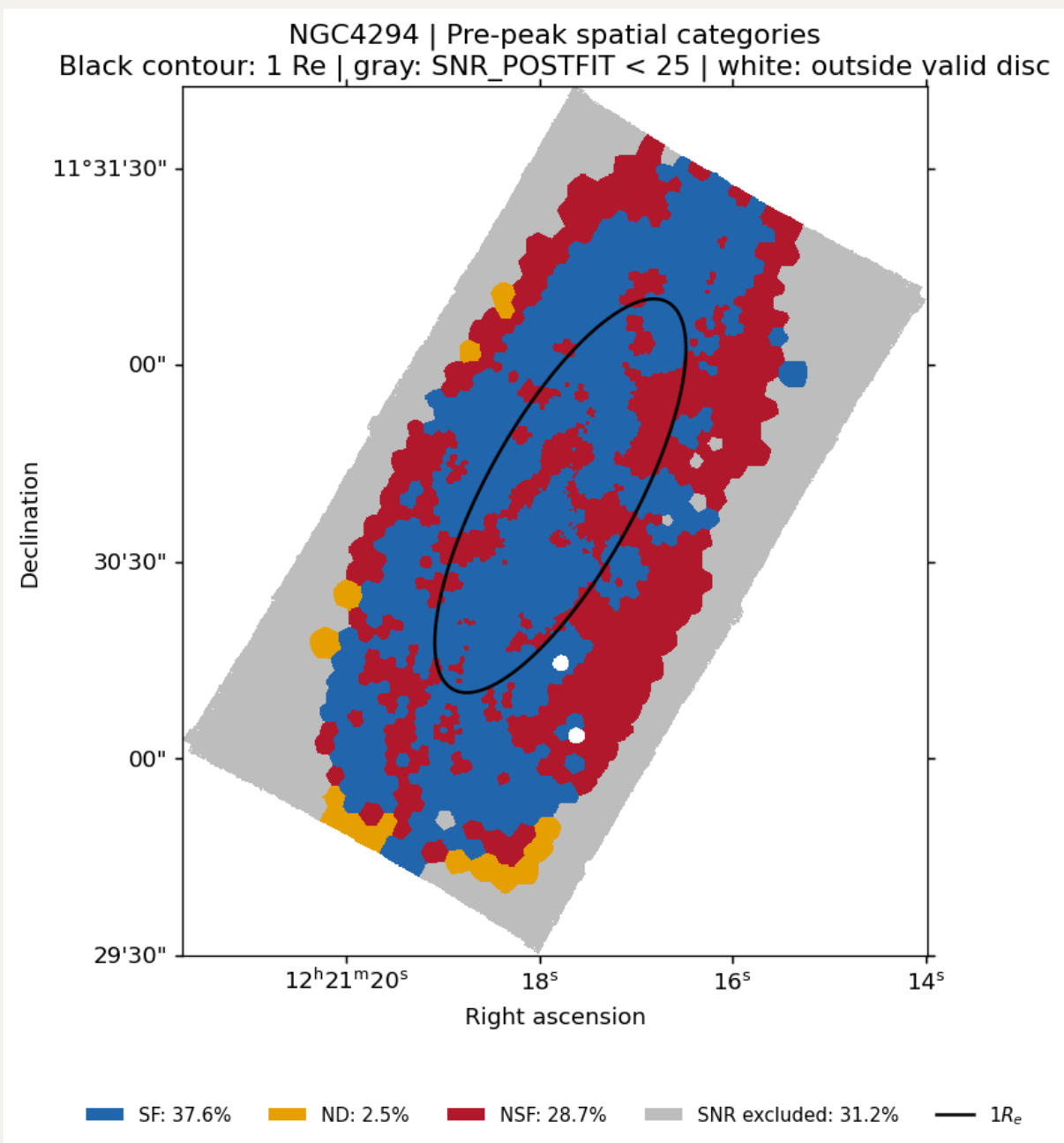
Individual Galaxy Spatial Map

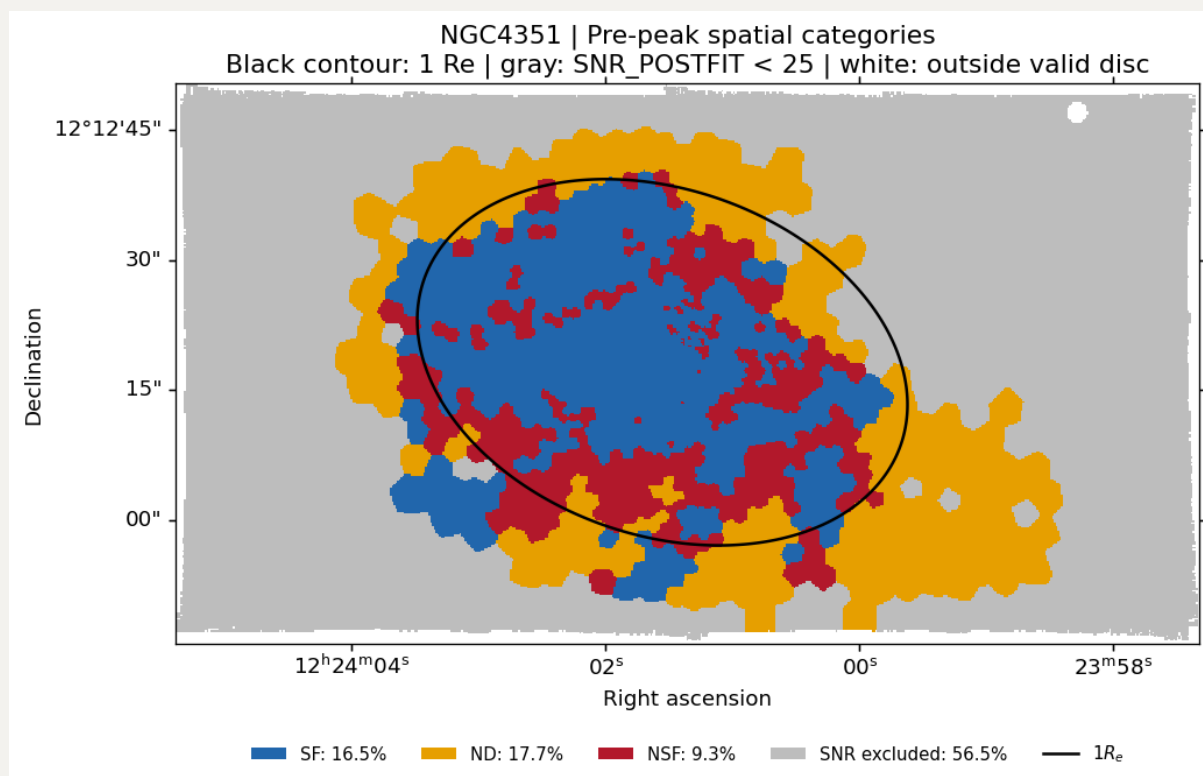
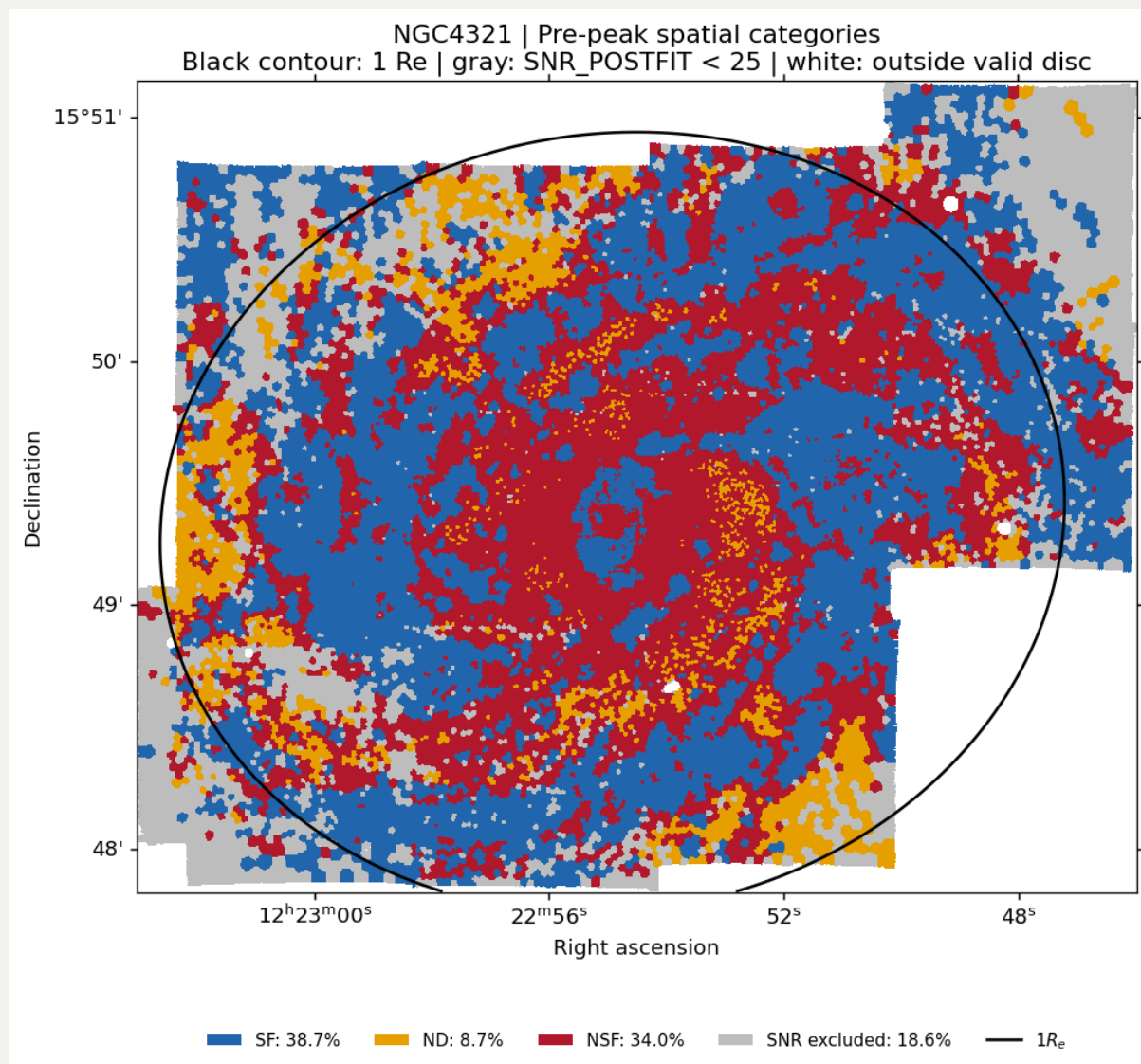


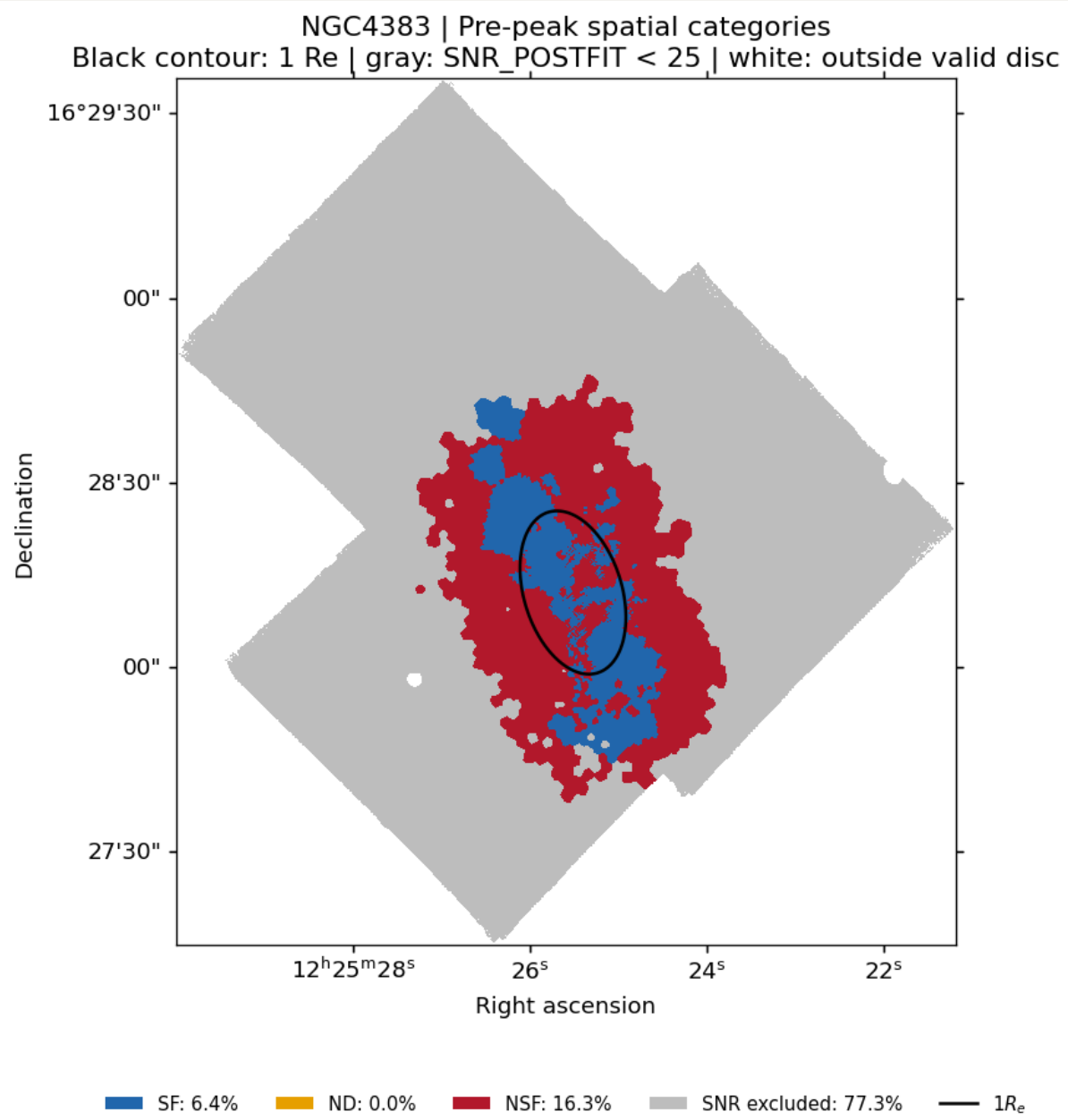
NGC4254 | Pre-peak spatial categories
Black contour: 1 Re | gray: SNR_POSTFIT < 25 | white: outside valid disc



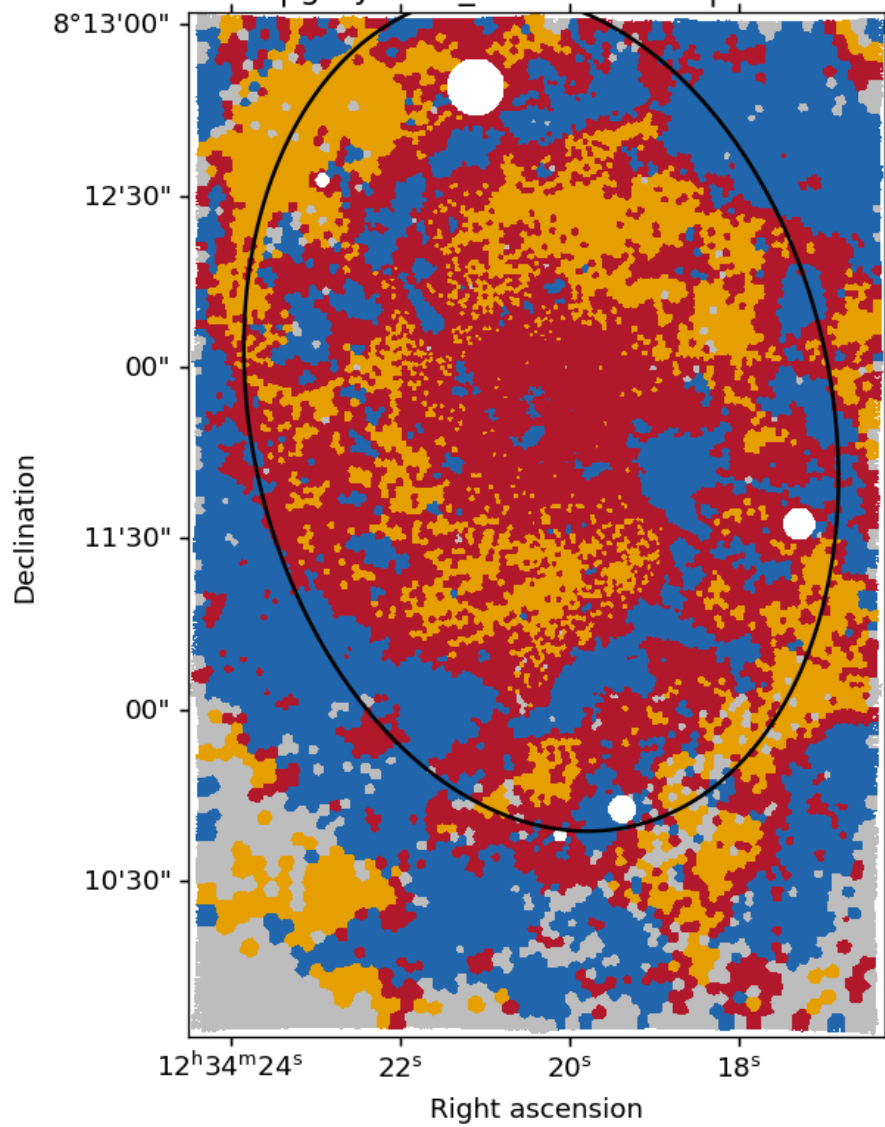
■ SF: 44.7% ■ ND: 0.8% ■ NSF: 15.7% ■ SNR excluded: 38.9% — 1Re



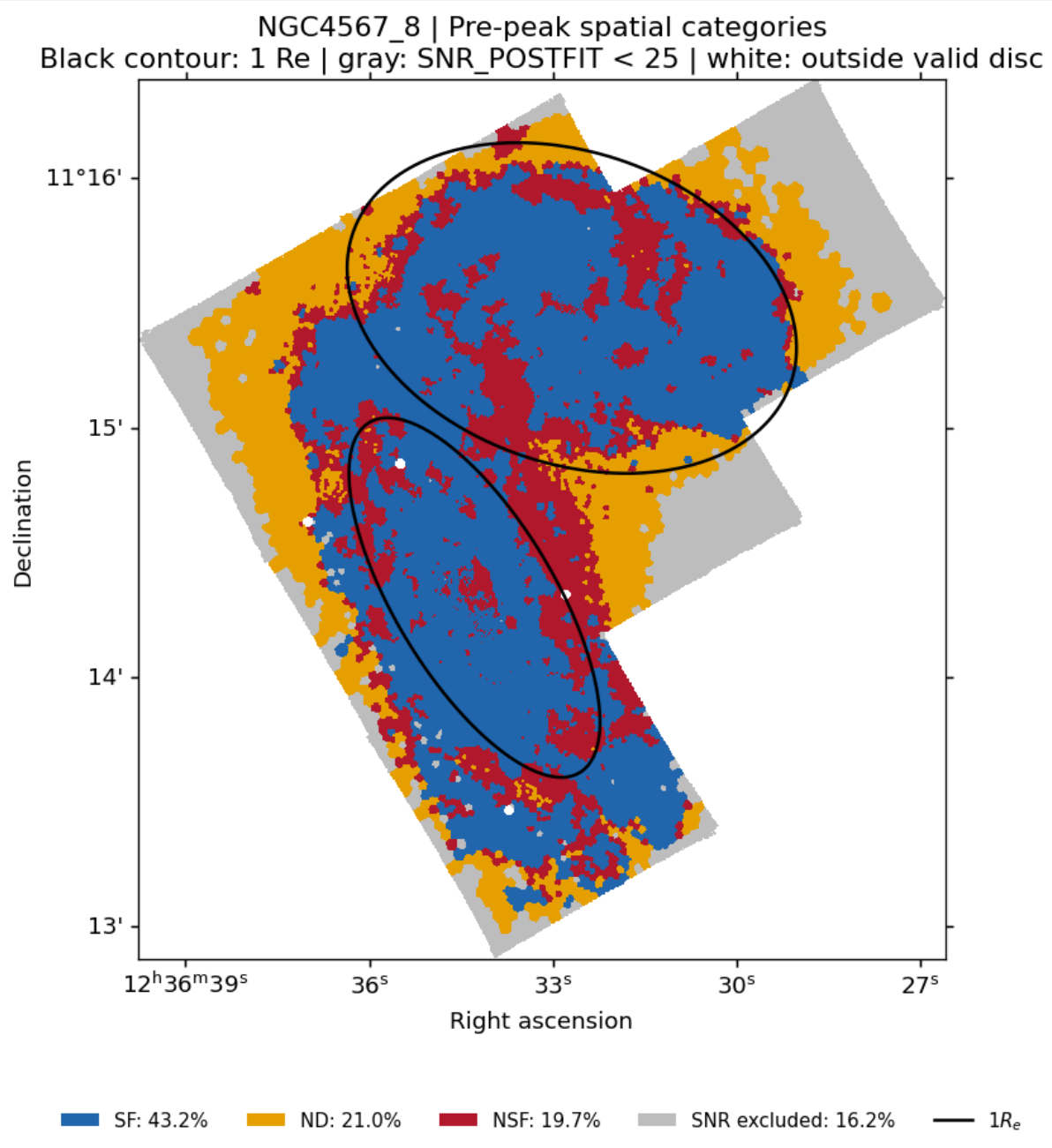




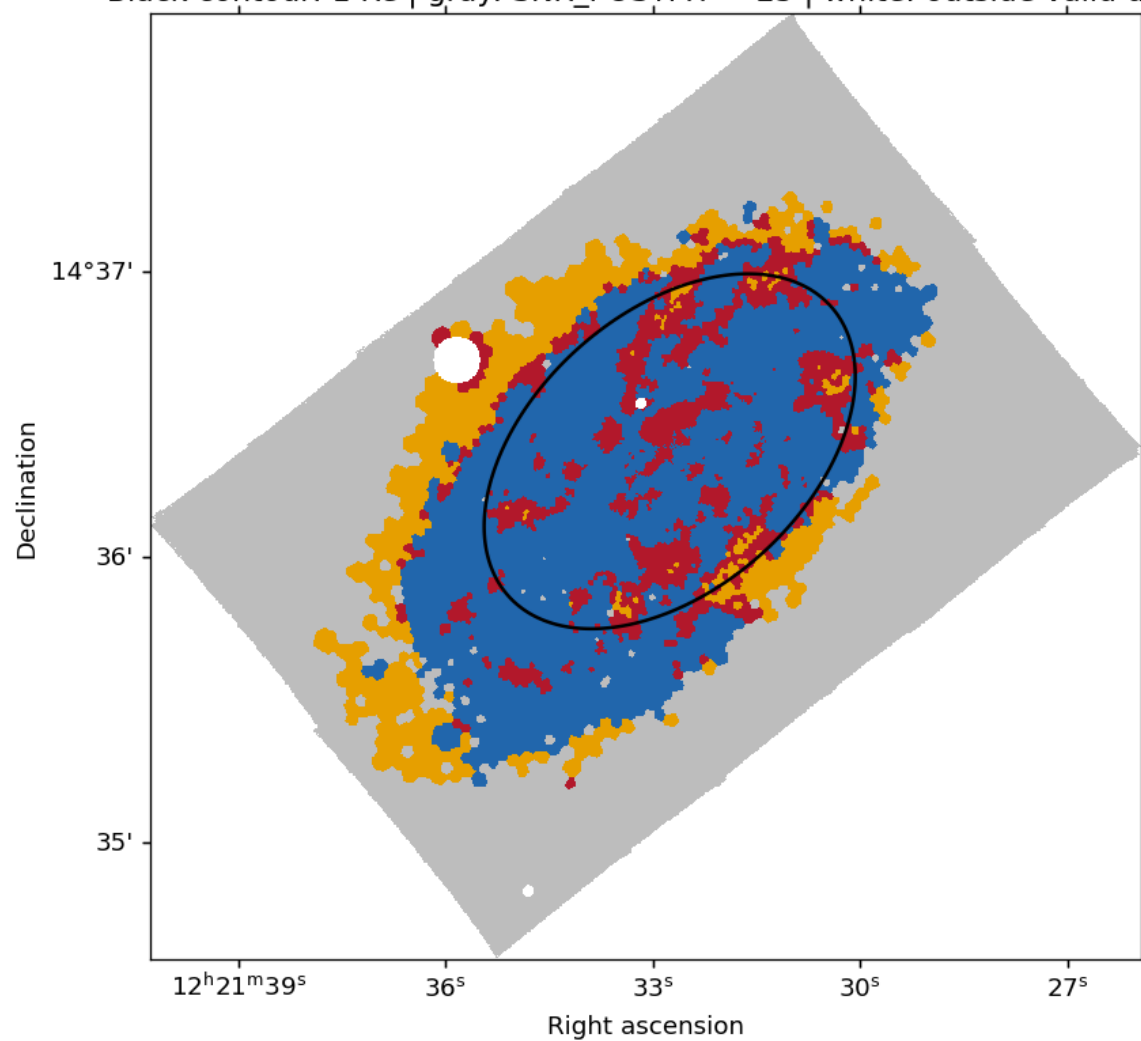
NGC4535 | Pre-peak spatial categories
Black contour: 1 Re | gray: SNR_POSTFIT < 25 | white: outside valid disc



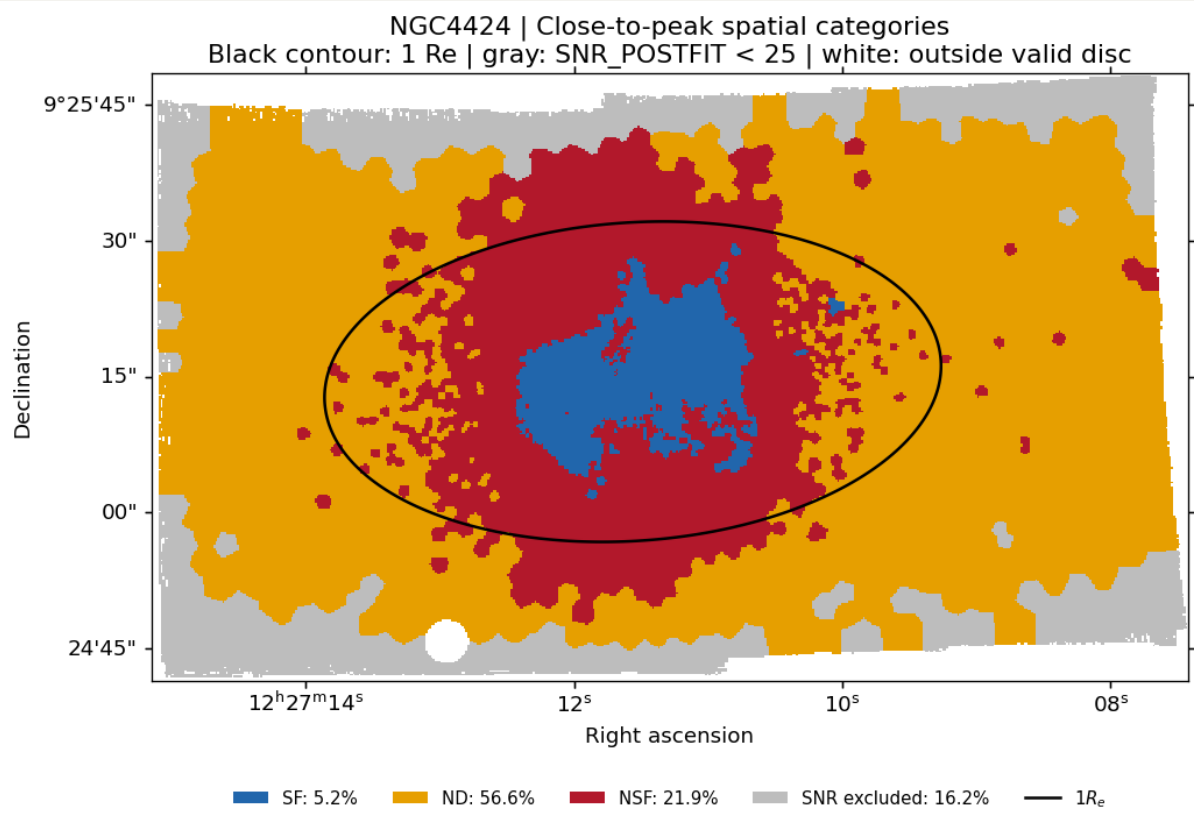
SF: 31.8% ND: 21.0% NSF: 37.4% SNR excluded: 9.7% — 1Re

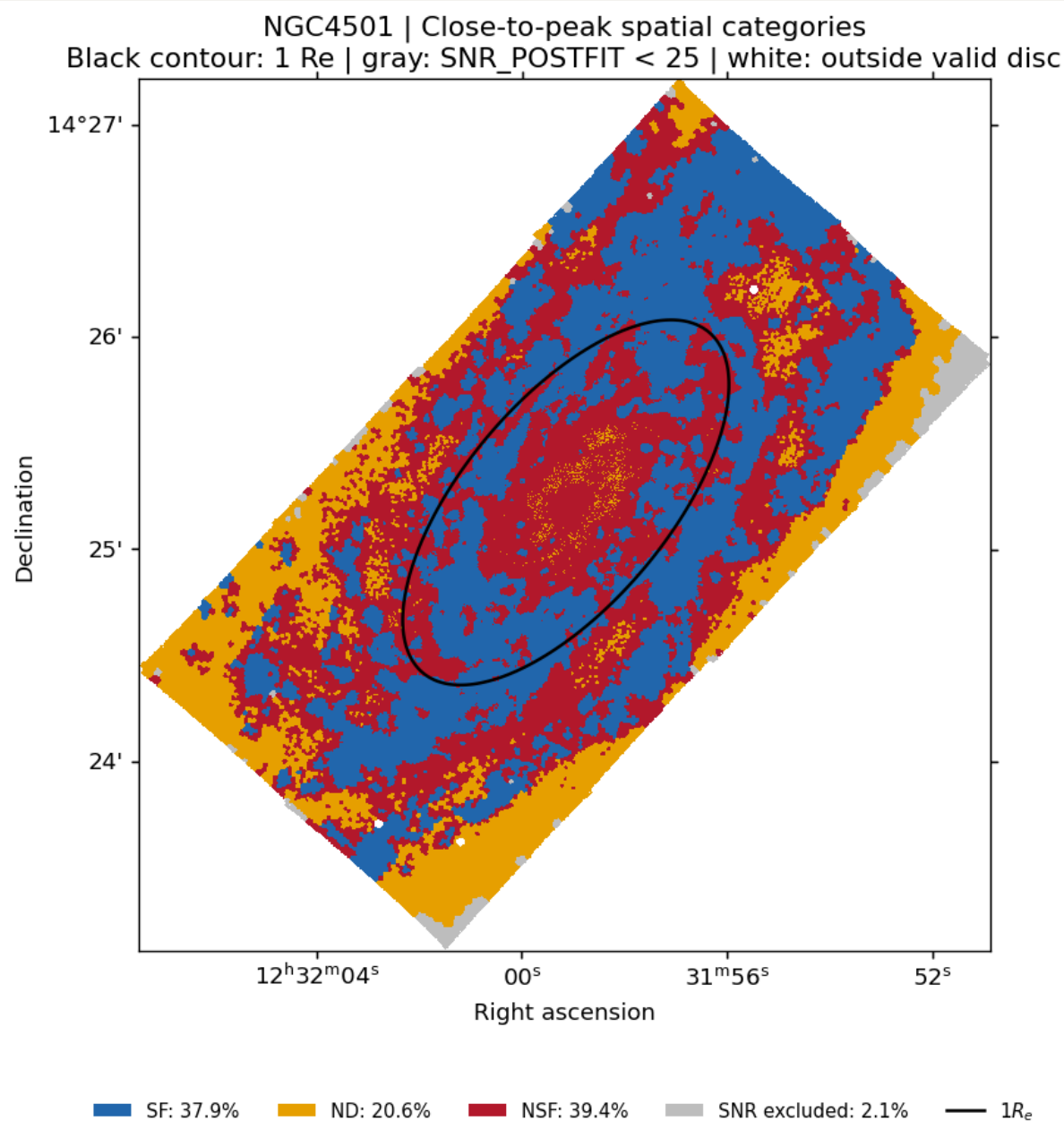


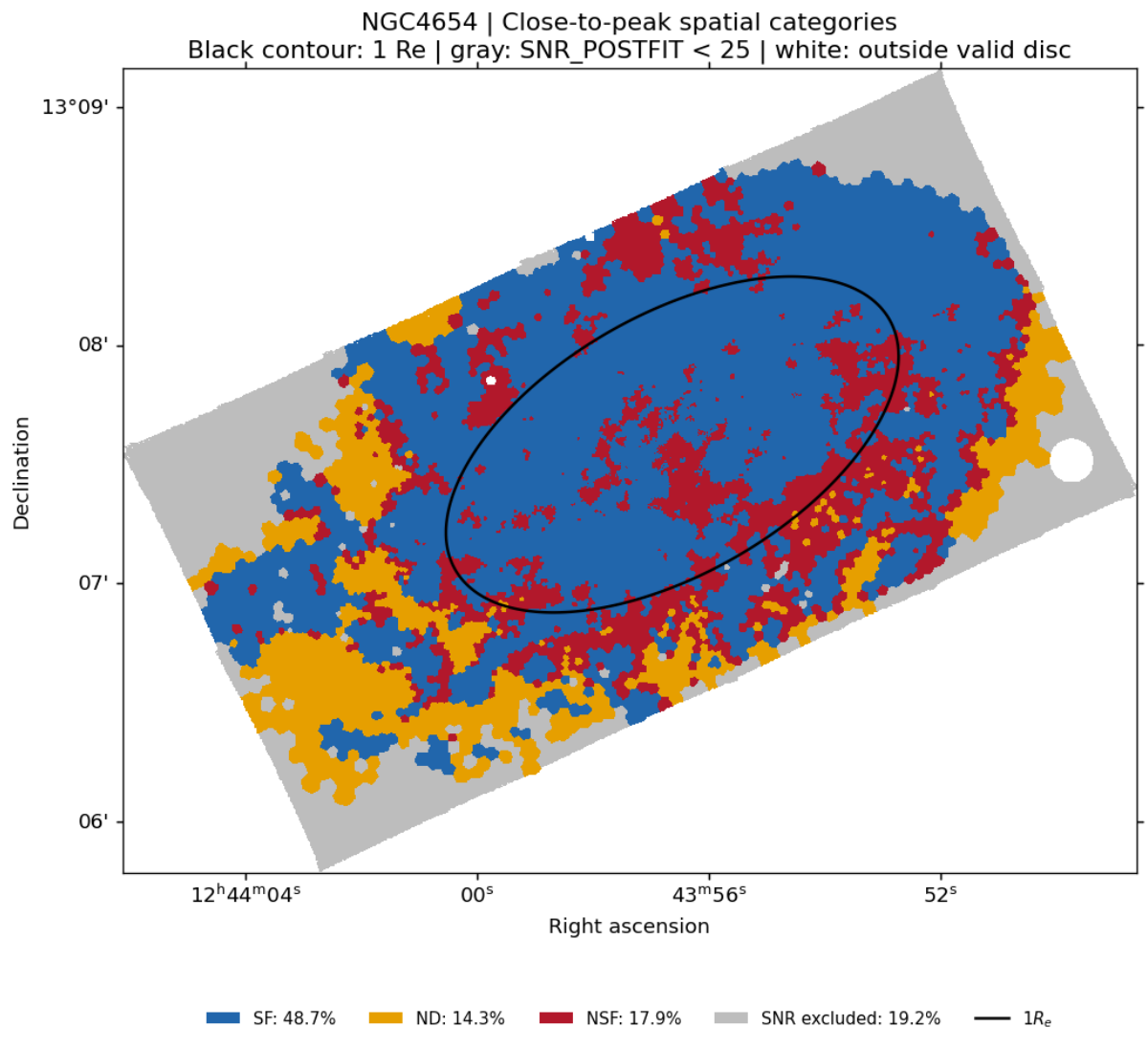
NGC4298 | Close-to-peak spatial categories
Black contour: 1 Re | gray: SNR_POSTFIT < 25 | white: outside valid disc

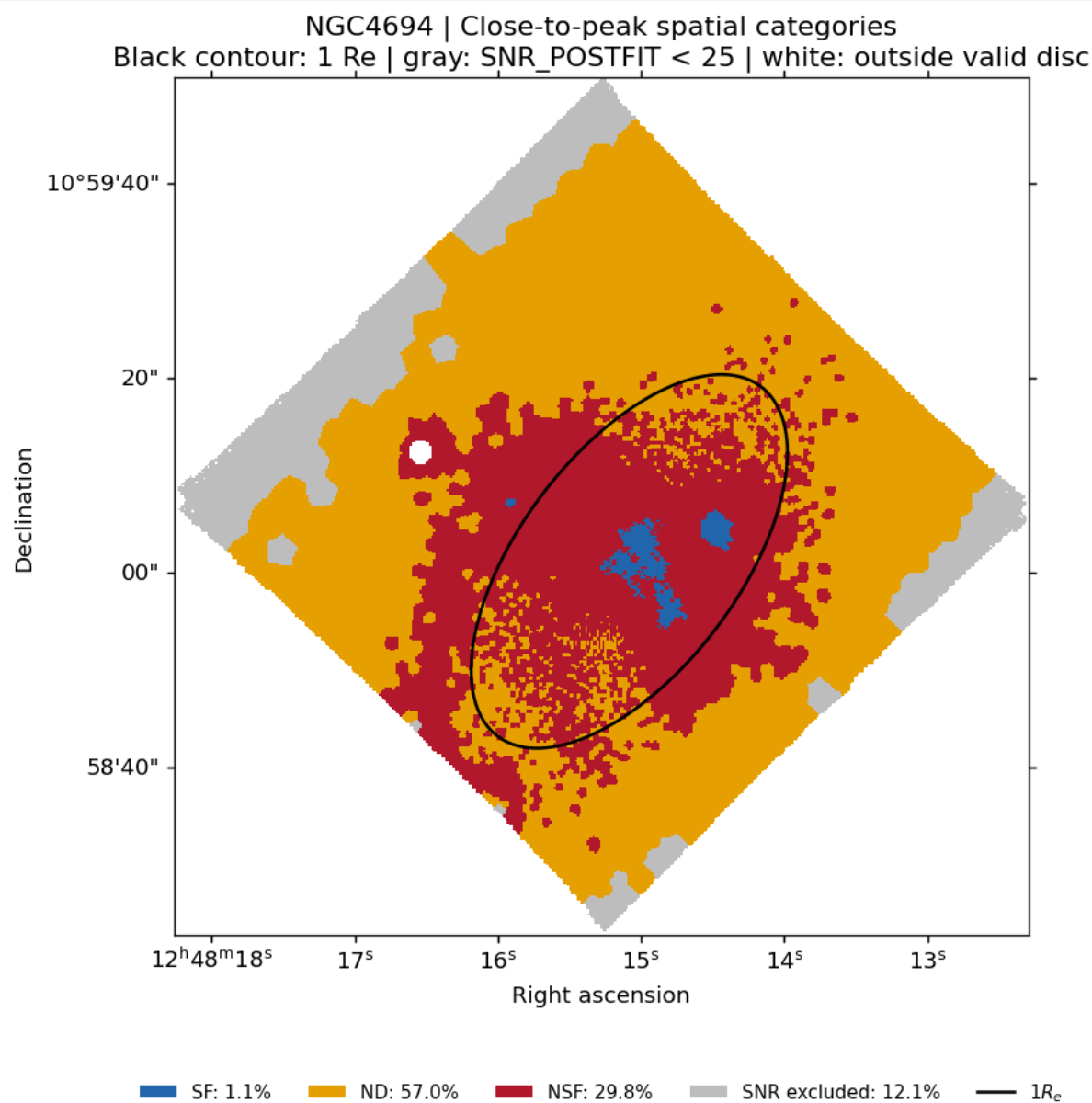


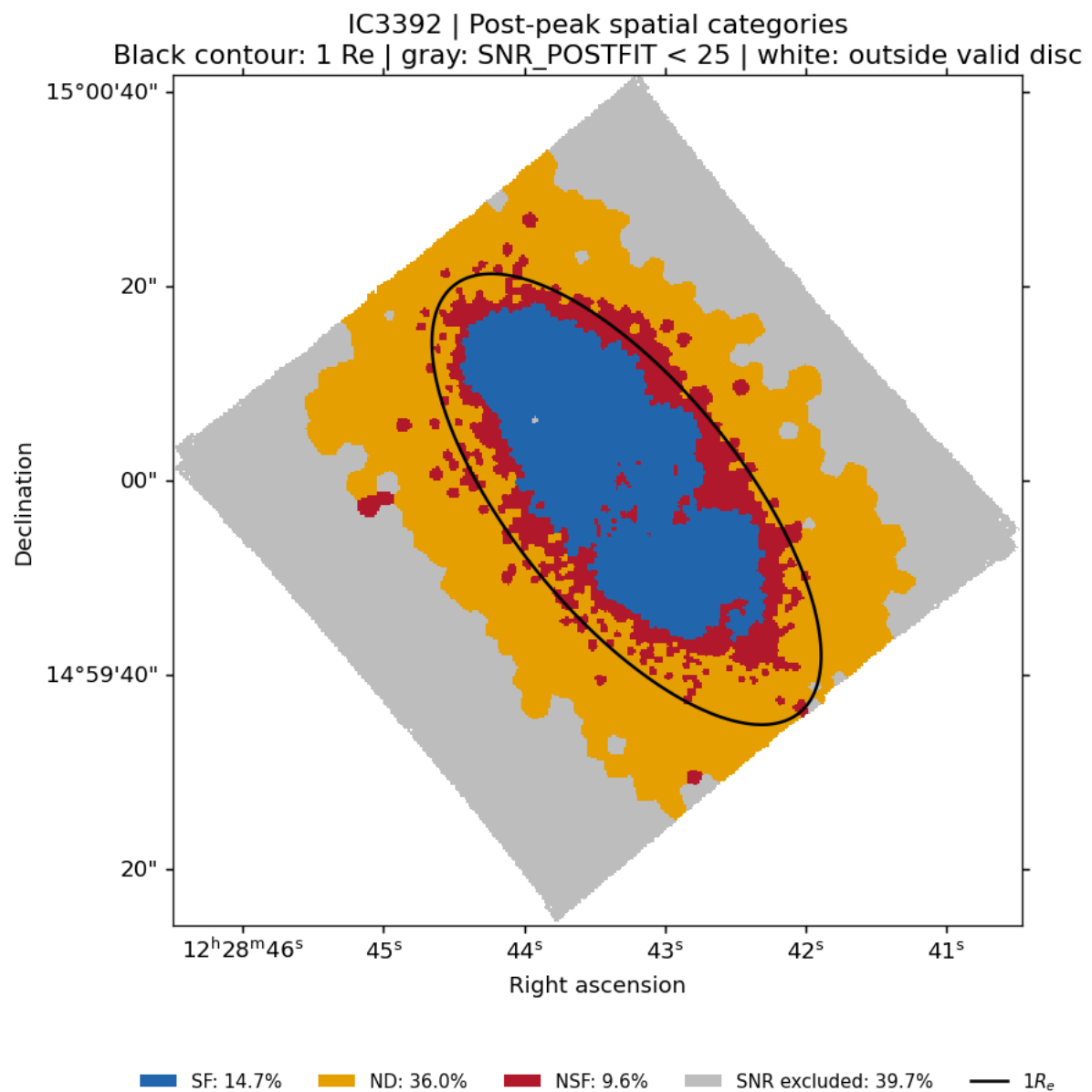
SF: 27.8% ND: 10.3% NSF: 8.6% SNR excluded: 53.3% 1Re

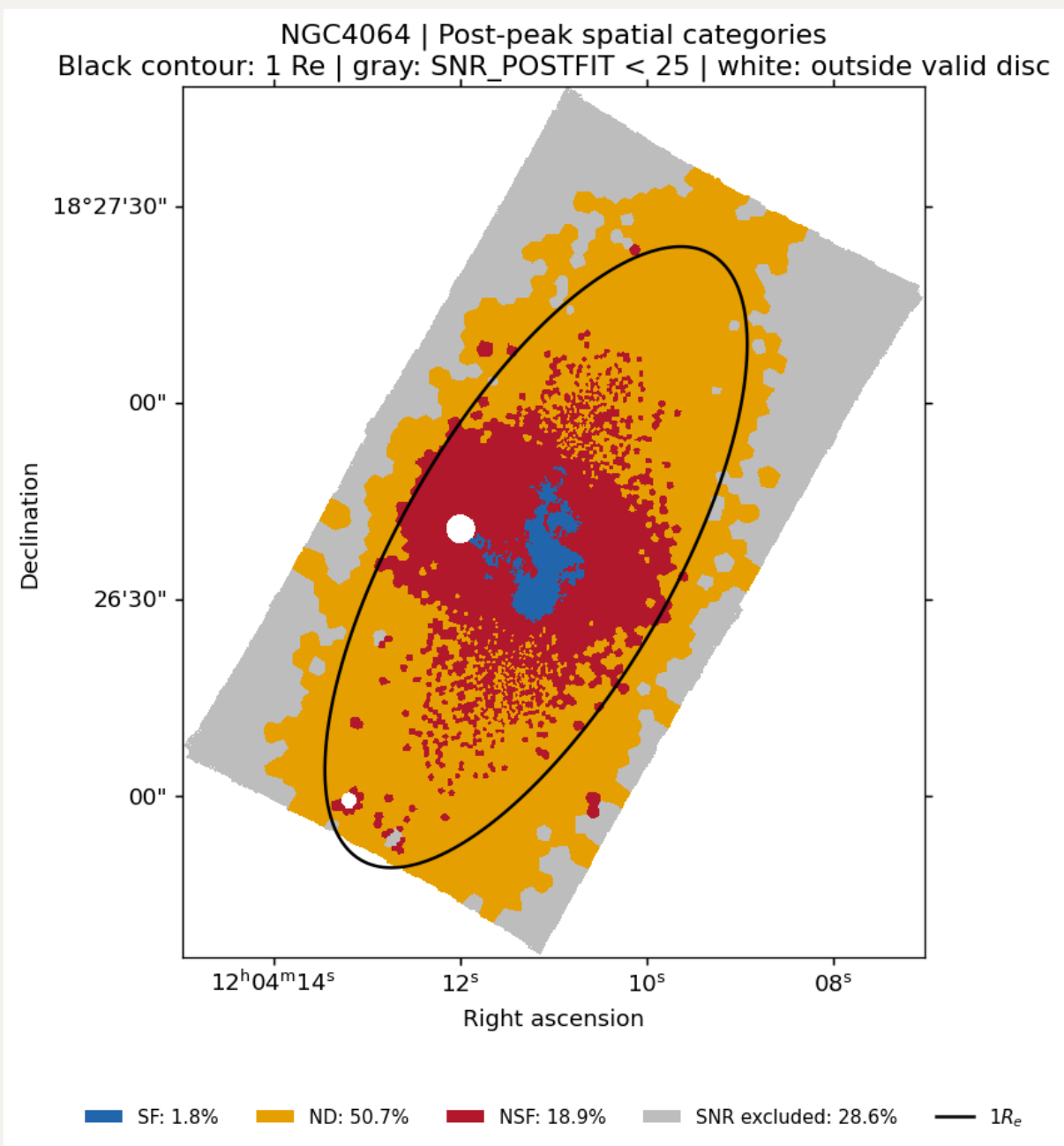


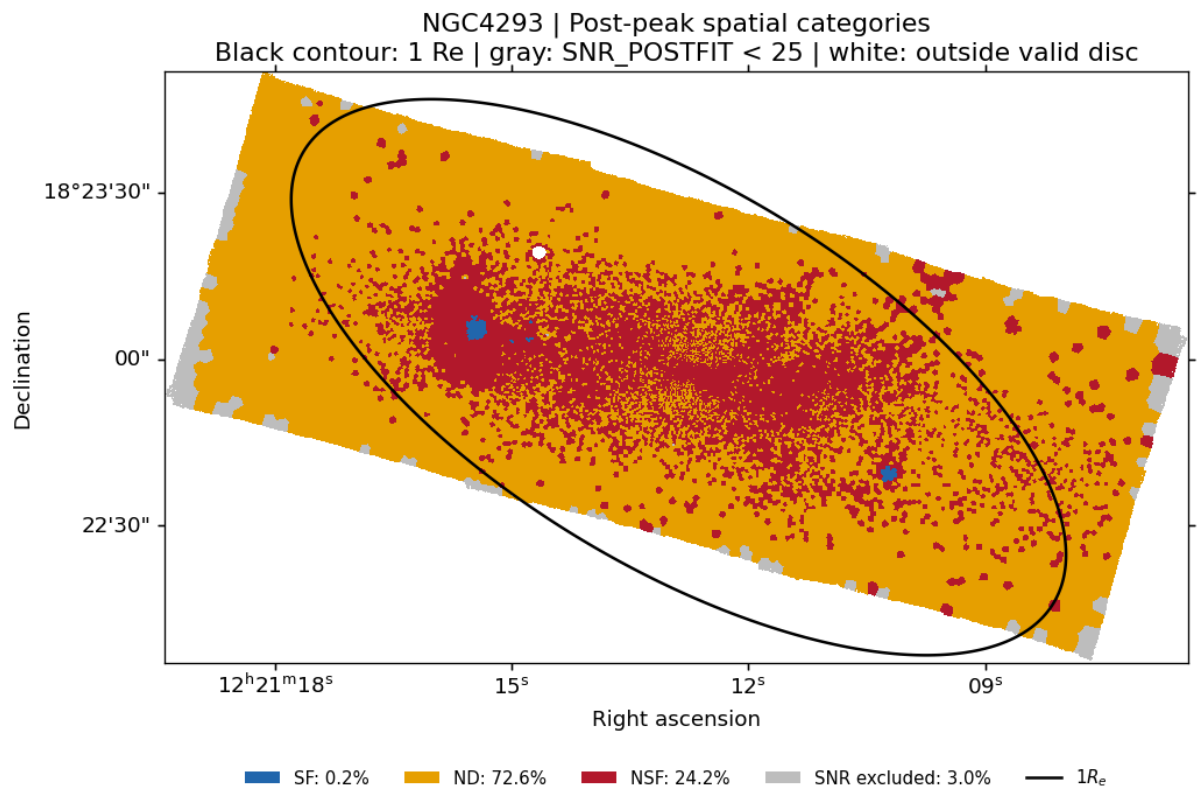


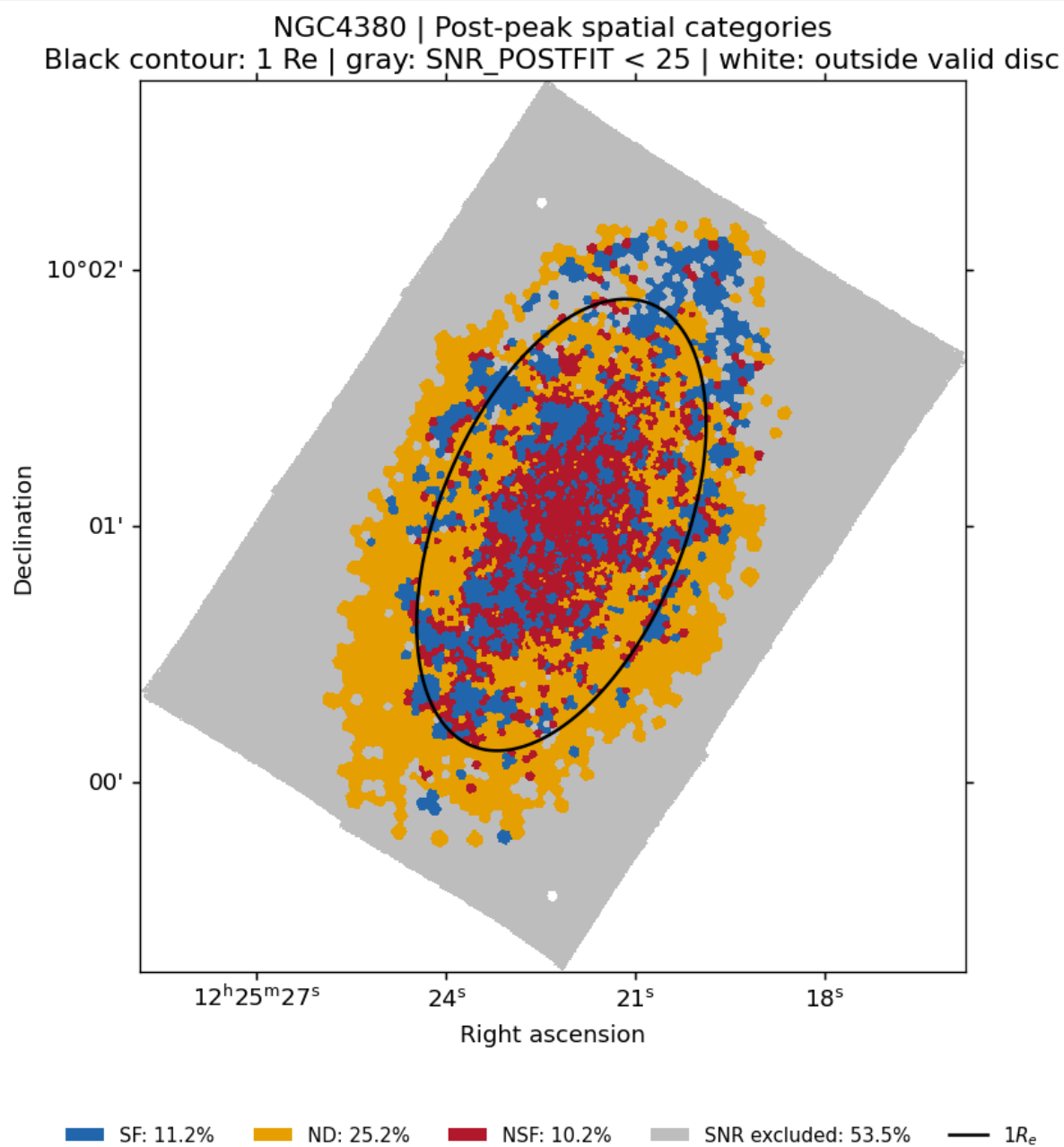




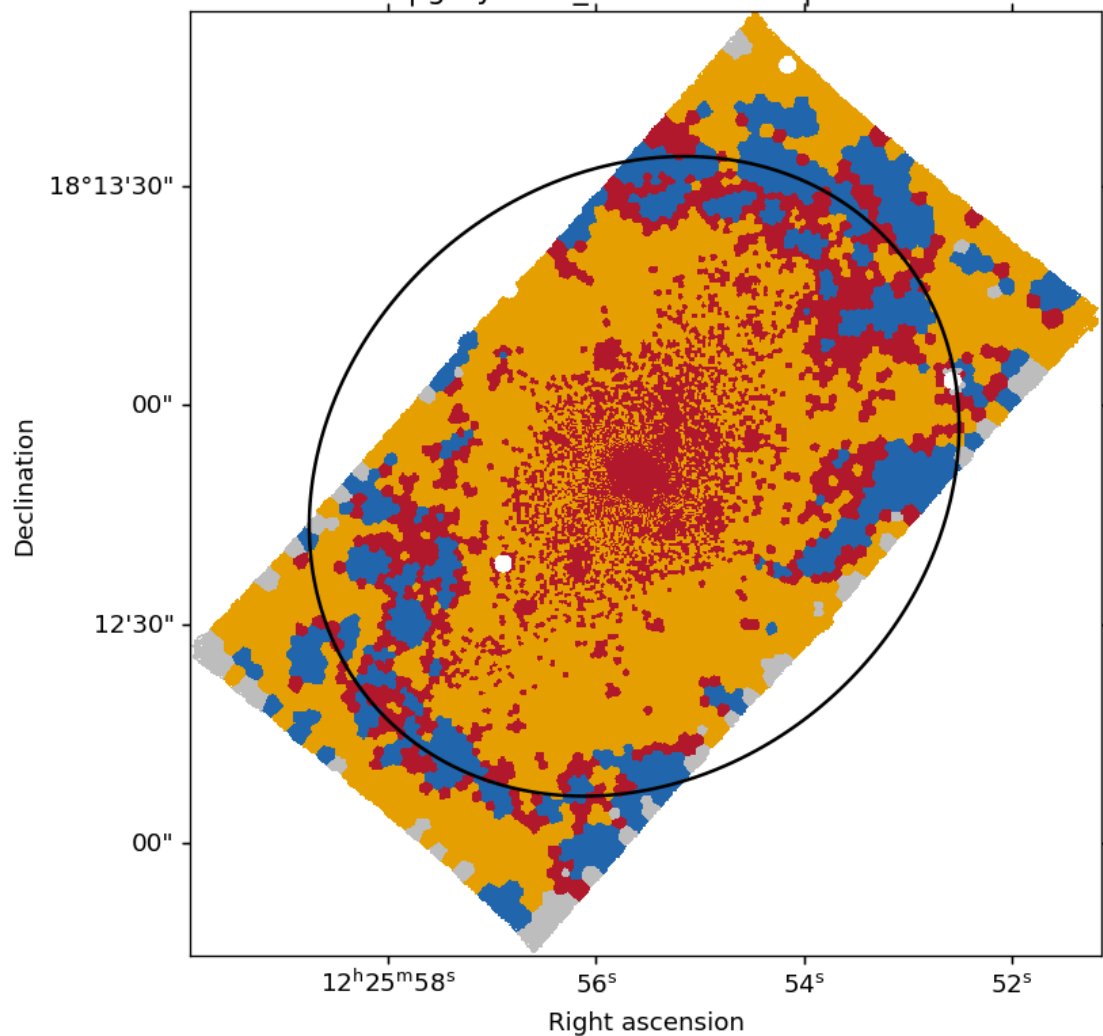






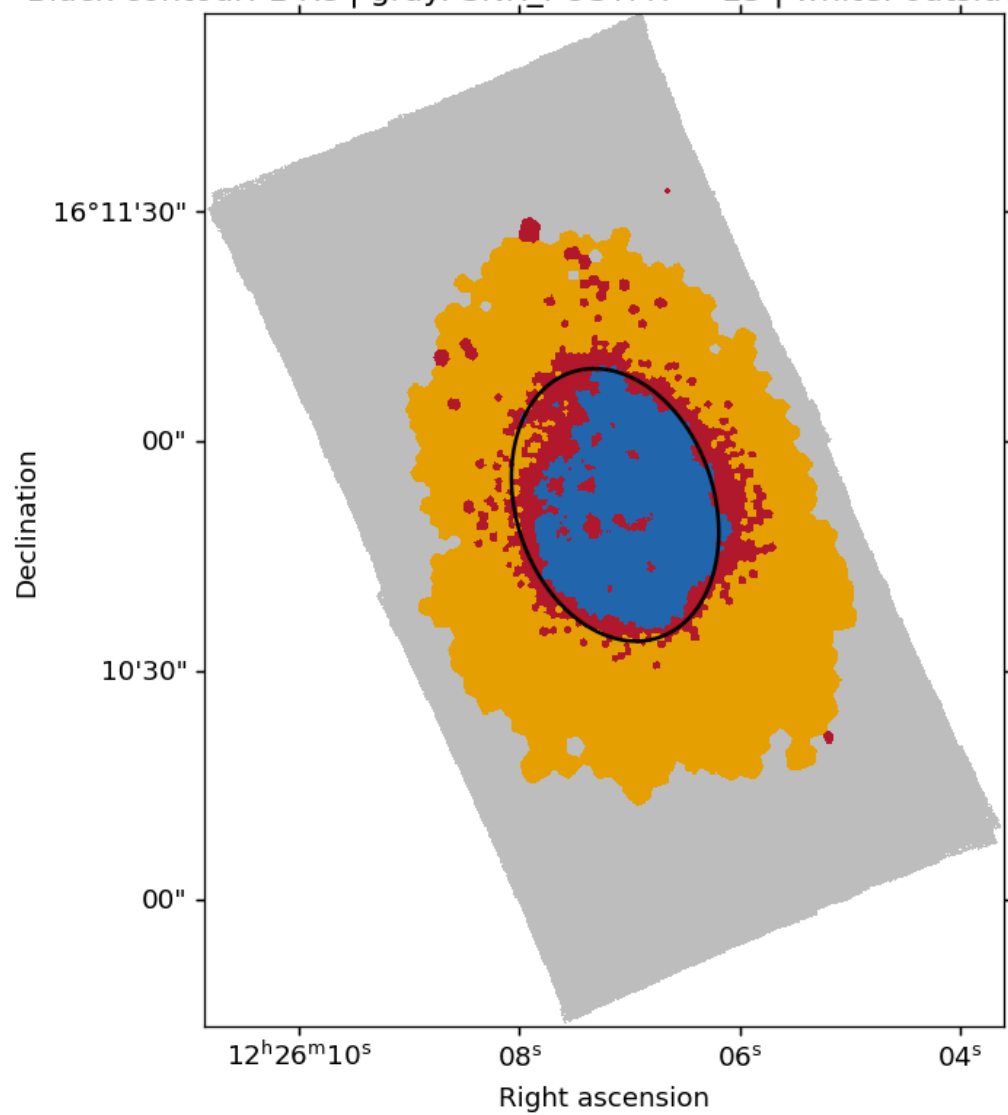


NGC4394 | Post-peak spatial categories
Black contour: 1 Re | gray: SNR_POSTFIT < 25 | white: outside valid disc

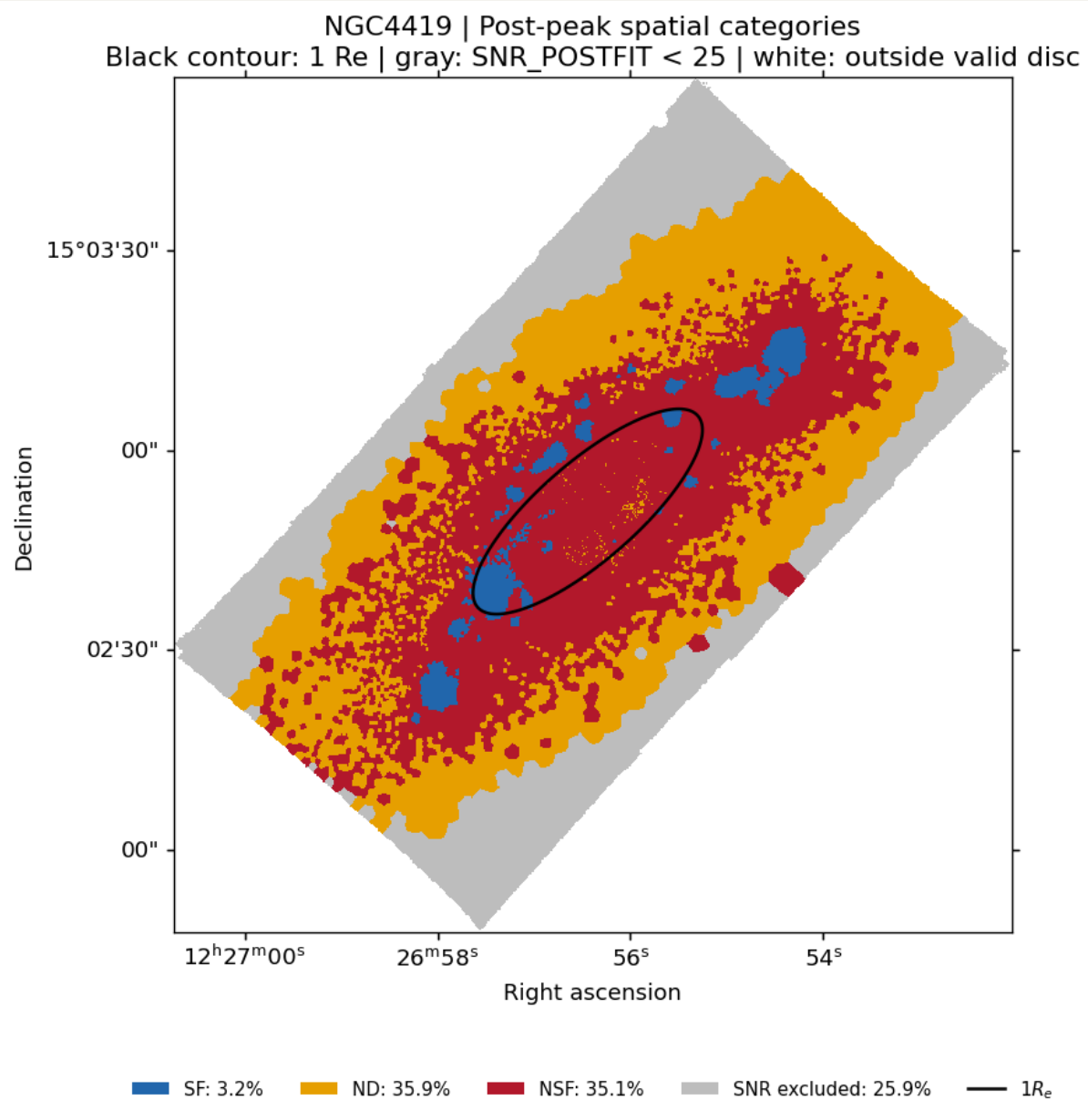


SF: 16.8% ND: 54.8% NSF: 25.1% SNR excluded: 3.3% — 1Re

NGC4405 | Post-peak spatial categories
Black contour: $1 R_e$ | gray: SNR_POSTFIT < 25 | white: outside valid disc

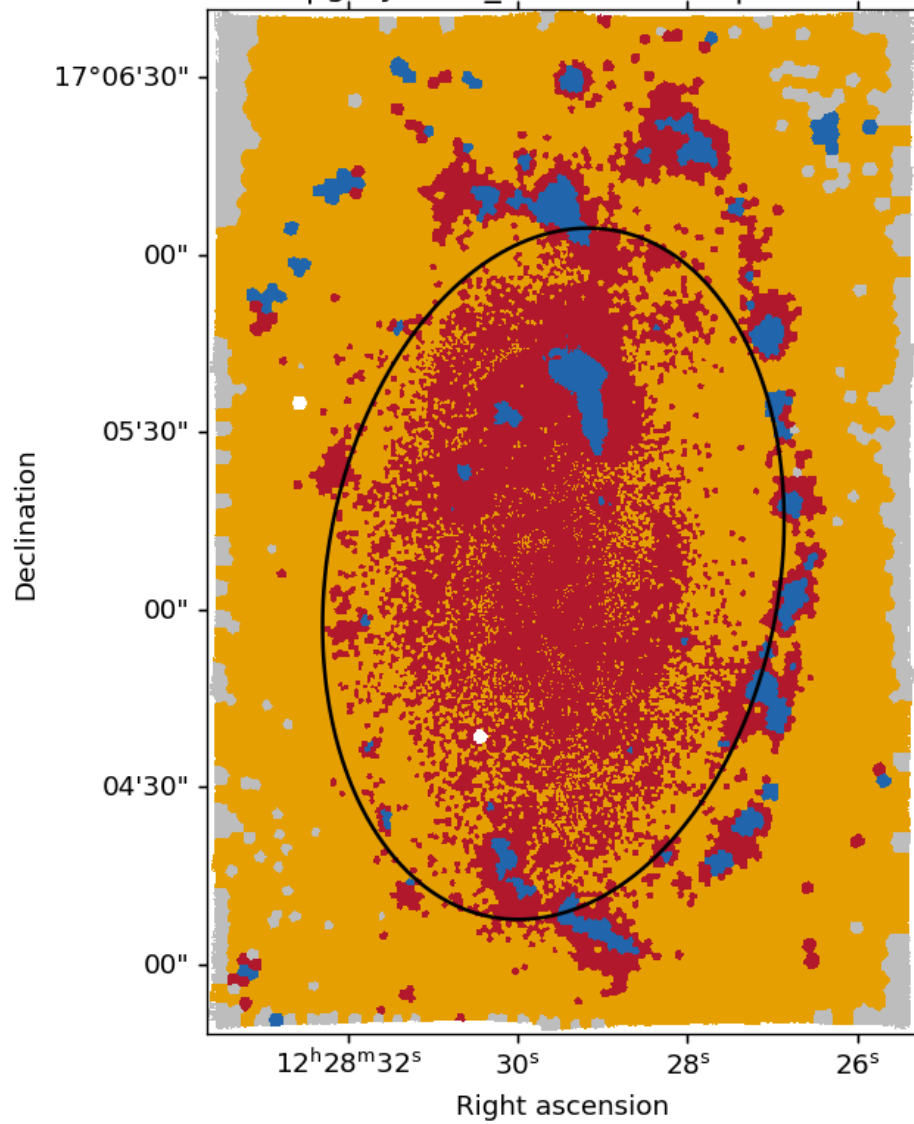


SF: 7.6% ND: 30.1% NSF: 6.4% SNR excluded: 55.9% $1 R_e$

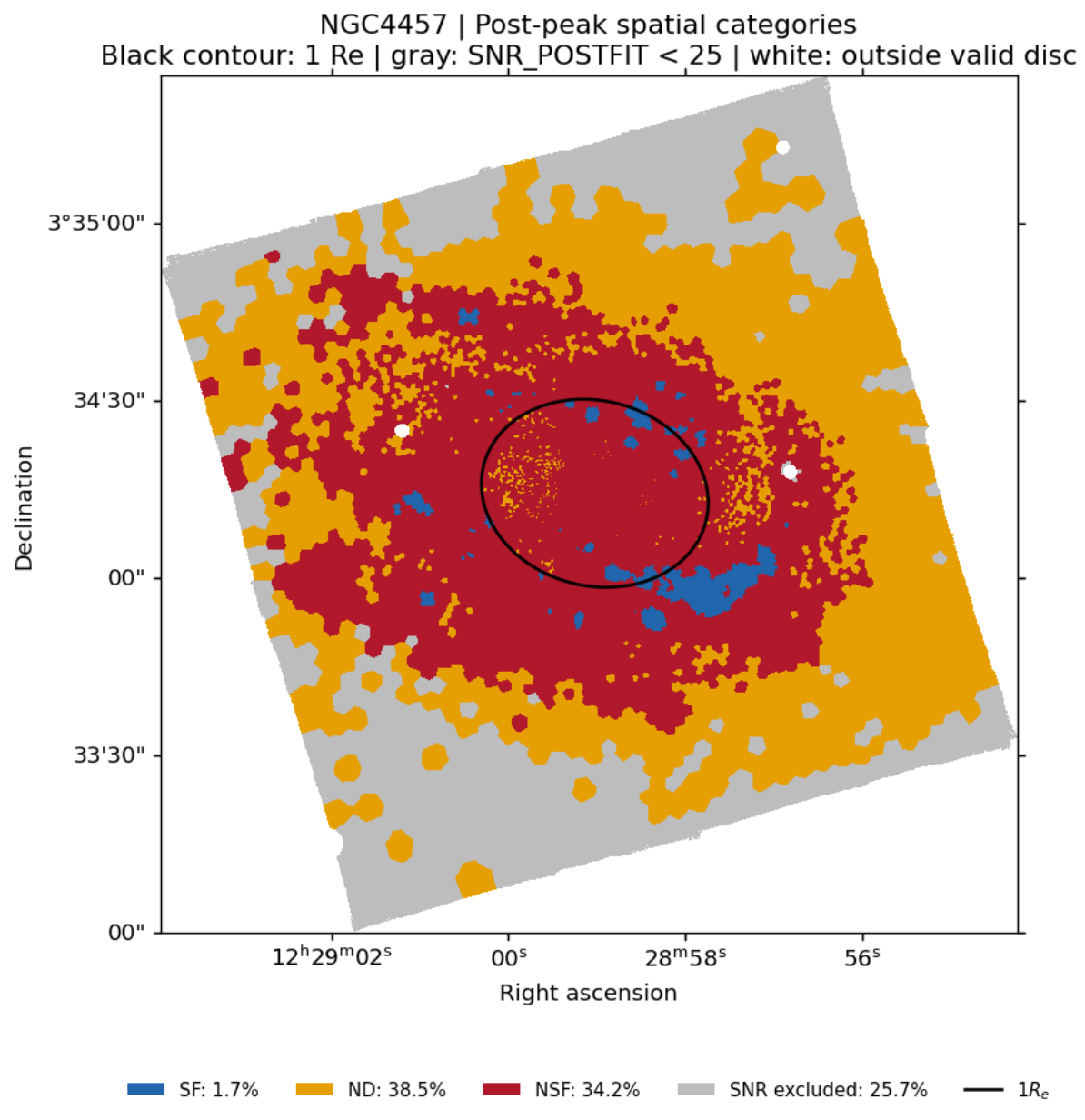


NGC4450 | Post-peak spatial categories

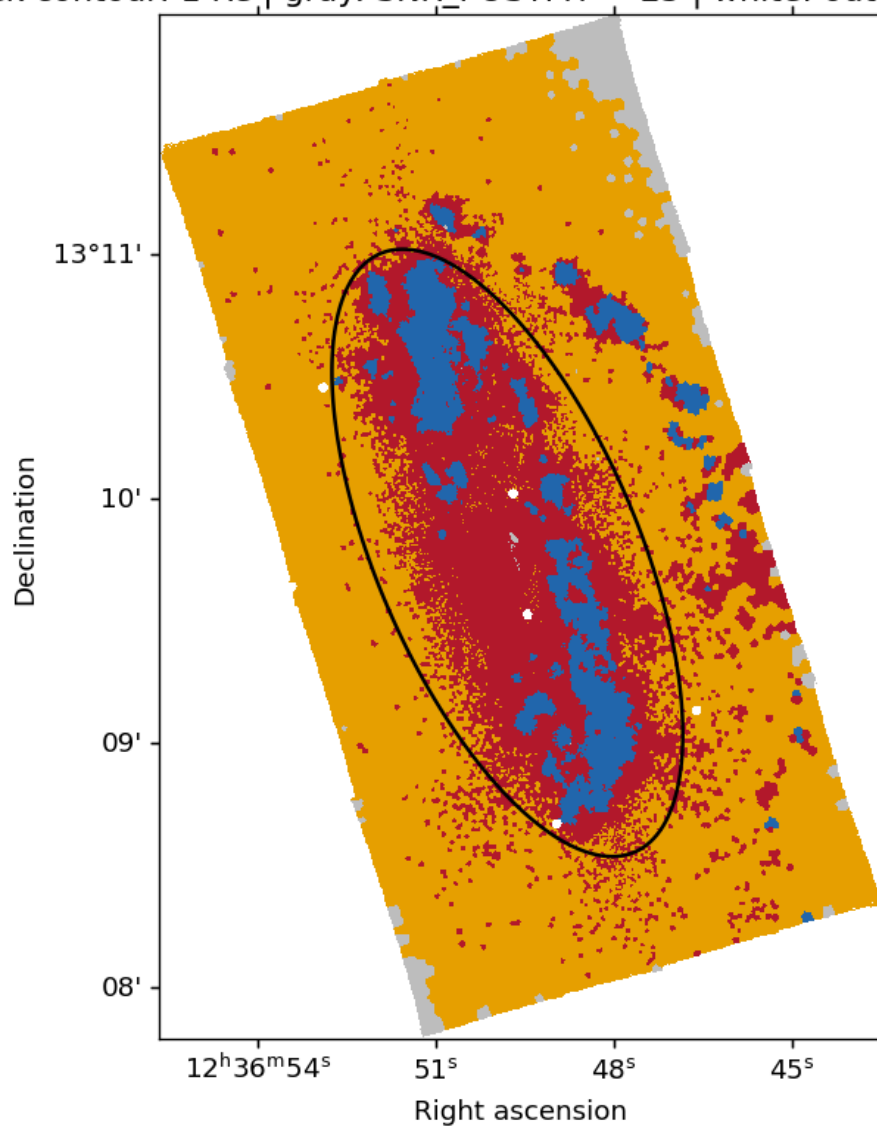
Black contour: 1 Re | gray: SNR_POSTFIT < 25 | white: outside valid disc



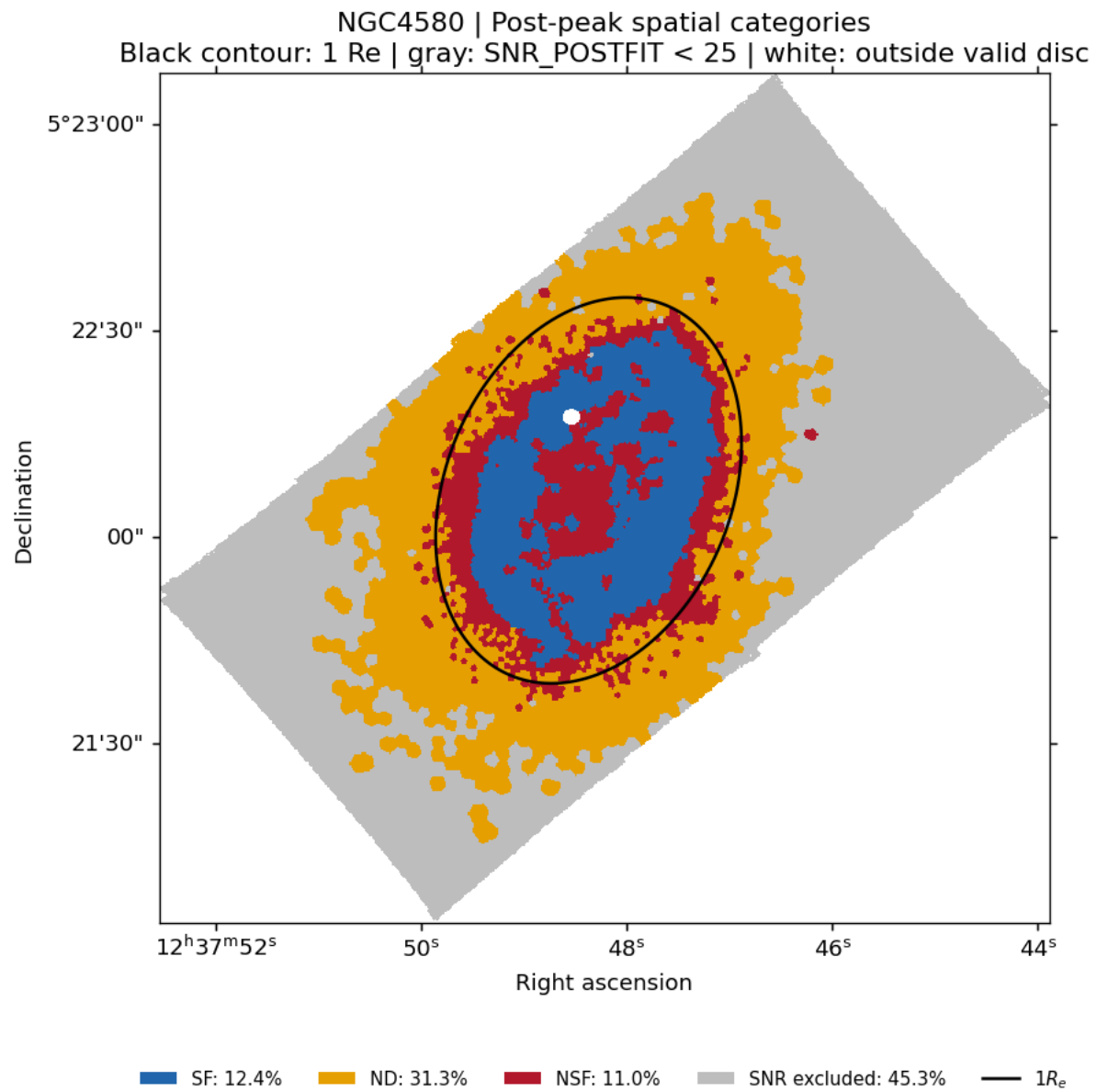
SF: 3.4% ND: 64.9% NSF: 25.7% SNR excluded: 5.9% — 1Re

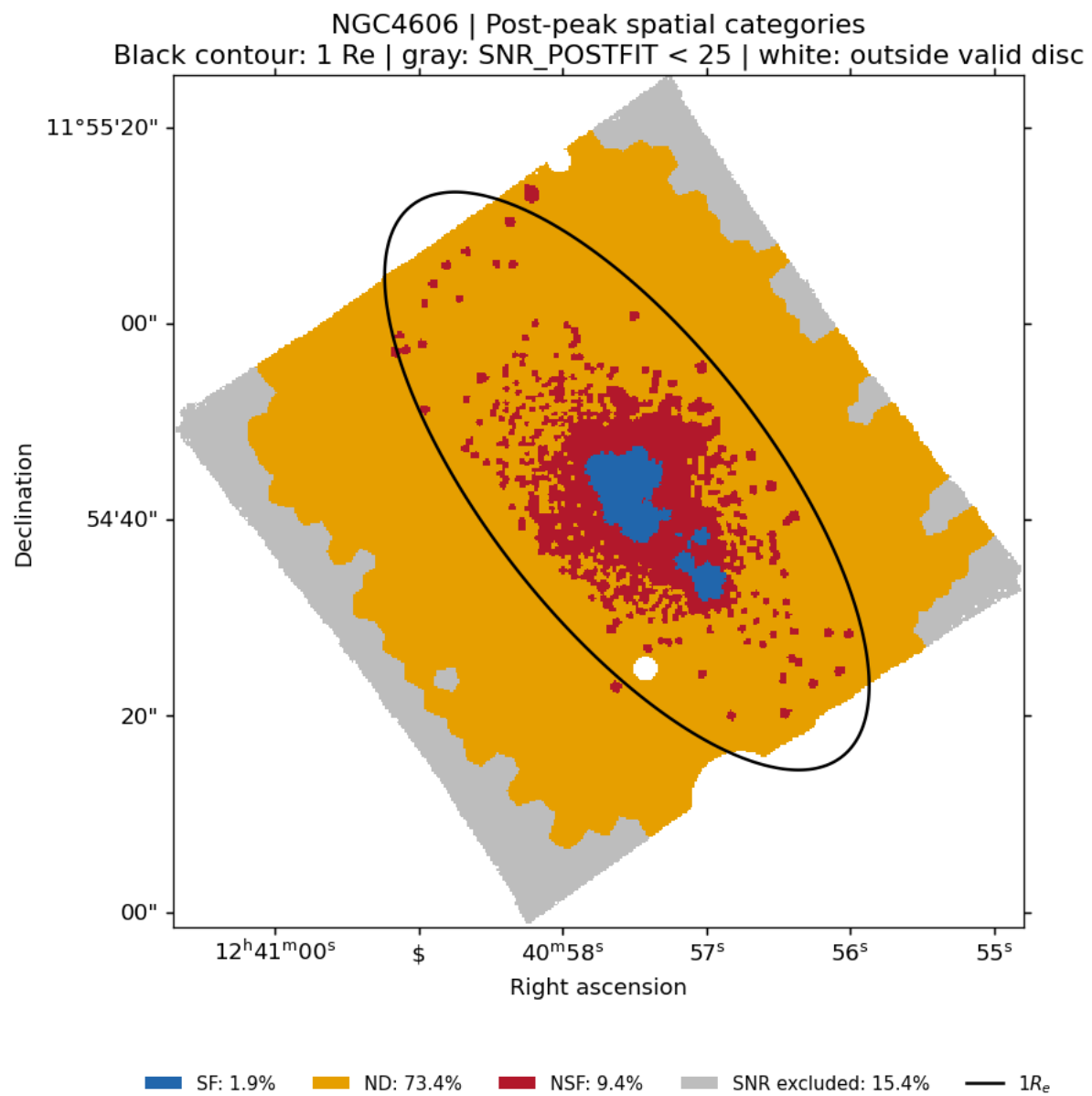


NGC4569 | Post-peak spatial categories
Black contour: 1 Re | gray: SNR_POSTFIT < 25 | white: outside valid disc



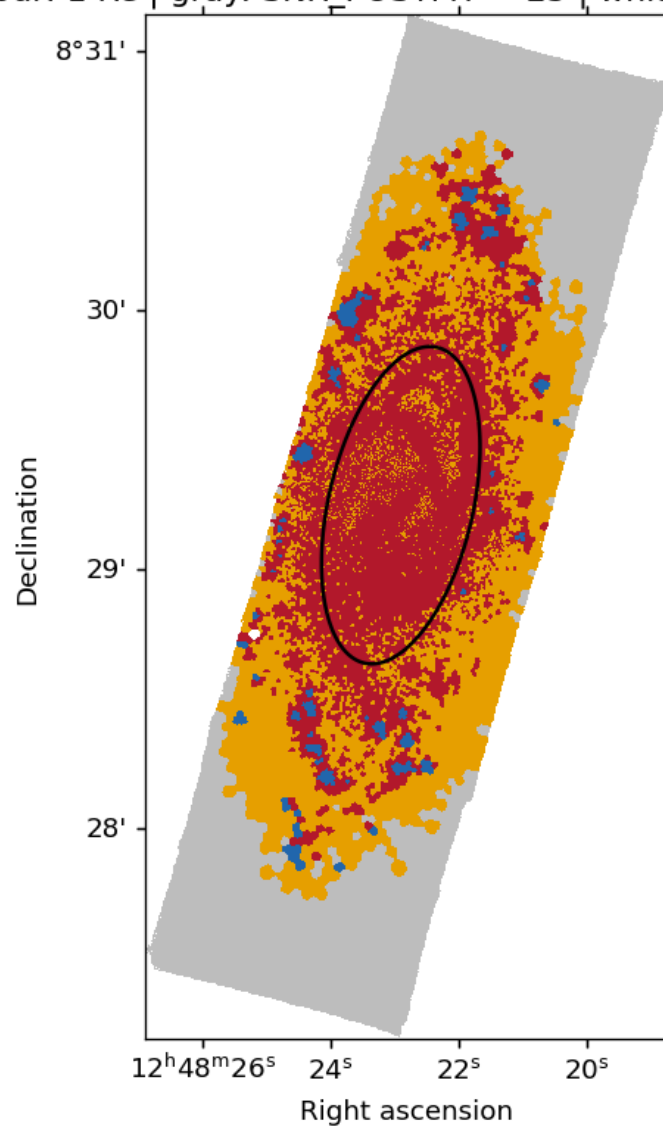
■ SF: 6.9% ■ ND: 65.7% ■ NSF: 24.5% ■ SNR excluded: 2.9% — 1R_e





NGC4698 | Post-peak spatial categories

Black contour: 1 Re | gray: SNR_POSTFIT < 25 | white: outside valid disc



SF: 1.9% ND: 32.2% NSF: 29.4% SNR excluded: 36.5% 1Re